

The aerodynamics of propellers

Quentin R. Wald*

102 Cape George Road, Port Townsend, WA 98368, USA

Abstract

The theory and the design of propellers of minimum induced loss is treated. The pioneer analysis of this problem was presented more than half a century ago by Theodorsen, but obscurities in his treatment and inaccuracies and limited coverage in his tables of the Goldstein circulation function for helicoidal vortex sheets have not been remedied until the present work which clarifies and extends his work. The inverse problem, the prediction of the performance of a given propeller of arbitrary form, is also treated. The theory of propellers of minimum energy loss is dependent on considerations of a regular helicoidal trailing vortex sheet; consequently, a more detailed discussion of the dynamics of vortex sheets and the consequences of their instability and roll up is presented than is usually found in treatments of propeller aerodynamics. Complete and accurate tables of the circulation function are presented. Interference effects between a fuselage or a nacelle and the propeller are considered. The regimes of propeller, vortex ring, and windmill operation are characterized.

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*Tel.: +1 360 379 6848.

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1. Introduction

1.1. Present status of propeller aerodynamics

The literature of propeller aerodynamics is scattered and in some respects is inconsistent and incomplete. Of basic importance for the theory and design of propellers is the treatment of propellers with load distribution for best efficiency developed by Theodorsen in a series of NACA reports and finally presented in his book published in 1948, but now long out of print. This work is a milestone in the development of the theory of propellers, but parts of it are obscure, it is not without errors, and the application to the design of an efficient propeller needs clarification. The consequence of these difficulties has been a general neglect, both in theoretical studies and in practical propeller design, of the underlying theory developed by Theodorsen. Though there have been several papers elucidating some aspects of this work, a

thorough reconsideration of it is lacking. It is hoped that the present study will bring attention to the deeper understanding of propeller aerodynamics.

It has been shown that the ideal distribution of circulation first computed by Goldstein need not be limited to the condition of light loading as assumed by Goldstein. Nonetheless propeller design methods in current use are limited by the light loading assumption and fail to take advantage of the more general possibilities.

Quite remarkable is the lack in the aeronautical literature of a complete and accurate tabulation of the Goldstein circulation function, which is essential for the design of a propeller with minimum energy loss. In general, it is a very difficult problem to compute rigorously the velocity induced by the vortex system. Theodorsen, employing an electrical analog, expanded Goldstein’s very limited tables, but his results are also limited and not very accurate. It is surprising that the David Taylor Model Basin published an accurate and

Nomenclature

a_0	lift slope, $dc_1/d\alpha$
B	number of blades
C_P	$P/\rho n^3 D^5$ power coefficient
C_Q	$Q/\rho n^2 D^5$ torque coefficient
C_T	$T/\rho n^2 D^4$ thrust coefficient
c	blade chord
c_d	section drag coefficient
c_l	section lift coefficient
D	propeller diameter
F	Prandtl's circulation correction, an approximation to $K(r, \lambda_2)$
$G(r, \lambda_2)$	Goldstein circulation function $\Gamma(r)/hw = B\Gamma\Omega/2\pi w(V+w)$
h	axial distance between adjacent turns of the vortex sheets, $2\pi(V+w)/\Omega B$
$K(r, \lambda_2)$	circulation function $\Gamma/\Gamma_\infty = [(x^2 + \lambda_2^2)/x^2]G$
K_P	$P/\frac{1}{2}\rho\pi R^2 V^3$ power coefficient
K_Q	$Q/\frac{1}{2}\rho\pi R^3 V^2$ torque coefficient
K_T	$T/\frac{1}{2}\rho\pi R^2 V^2$ thrust coefficient
K_{P1}	$P/\frac{1}{2}\rho\pi R_1^2 V^3$ power coefficient referred to the ideal vortex trail
K_{Q1}	$Q/\frac{1}{2}\rho\pi R_1^3 V^2$ torque coefficient referred to the ideal vortex trail
K_{T1}	$T/\frac{1}{2}\rho\pi R_1^2 V^2$ thrust coefficient referred to the ideal vortex trail
n	propeller revolutions/sec.
P	Power
p	static pressure
Q	Torque
R	radius of propeller
R_1	radius to edge of trailing vortex sheets
r	general radial coordinate
r_0	radial coordinate at propeller disc
r_1	radial coordinate in trailing vortex system
S	control surface; also area of axial projection of trailing vortex system πR_1^2
T	thrust
U_0	resultant velocity at a blade element
u	velocity
$\bar{u}$	u/V

u_n	velocity normal to vortex sheet
u_r	radial velocity
u_θ	tangential velocity
u_z	axial velocity
V	velocity of advance
w	axial displacement velocity of helical vortex sheets far behind the propeller
$\bar{w}$	w/V
x	r/R , dimensionless radial coordinate at the propeller
x_1	r/R_1 , dimensionless radial coordinate on the trailing vortex system
z	axial coordinate, downstream from propeller plane
α	blade angle of attack
α_{L_0}	angle of attack for zero lift
β	blade angle from plane of rotation
Γ	circulation
ε	axial kinetic energy factor $\int_S (u_z^2/w^2 S) dS$
η	efficiency
η_i	efficiency of ideally loaded frictionless propeller
θ	angular coordinate in the system r, θ, z
κ	mass transport coefficient
	$\int_S (u_z/w S) dS = \int_0^1 2G(x_1)x_1 dx_1$
λ	advance ratio $V/\Omega R$
λ_1	advance ratio $V/\Omega R_1$
λ_2	$(V+w)/\Omega R_1 = (1+\bar{w})\lambda_1$
ρ	density
σ	$Bc/2\pi R$, solidity factor
ϕ	$\tan^{-1}(\lambda_2/x)$, pitch angle of helical vortex sheet.
ϕ	velocity potential
Ω	angular velocity

Subscripts

0	at the propeller plane (usually omitted)
1	coefficients and variables referred to the helicoidal trailing vortex system
h	at the hub
t	at the tip
Bold face denotes a vector	

complete tabulation of a simply related function as long ago as 1964, and yet this work appears to be unknown to those concerned with aircraft propellers.

The purpose of this work is to present a re-examination of the theory of aircraft propellers with ideal load distribution, to clarify some of the more

obscure points of Theodorsen's treatment, to present a systematic design procedure, and to provide accurate and more complete tables of the circulation function and the mass coefficient that are necessary in the design process. The effect of a large hub or spinner on the circulation distribution has also been introduced. A method for the computation of the performance off design conditions and for arbitrary propellers is also presented.

The theory of aircraft propellers, following the original development of finite wing theory, has nearly always proceeded as a lifting line analysis. That is, blade elements may be considered to act as two-dimensional foils upon which the forces are the same as would be found in a uniform two-dimensional flow with the same velocity and direction as occurs locally at the blade element. This approach to the design of blade elements is continued in the present study. The lifting line treatment does not restrict the generality of the underlying analysis of the trailing vortex system.

Since the theory of propellers with minimum induced loss is founded on considerations of the trailing vortex sheet, it was thought to be necessary to present a more detailed discussion of the dynamics of vortex sheets and the consequences of their instability and roll up than is usually found in treatments of propeller aerodynamics.

1.2. Historical development of propeller theory

The development of a rational theory of propeller action begins with the work of Rankine [1] and Froude [2] in the 19th century. Their interest was in marine propulsion, but the fundamental principles are, of course, the same for water and air. They developed the fundamental momentum relations governing a propulsive device in a fluid medium. At the end of the century Drzewiecki [3] presented a theory of propeller action where blade elements were treated as individual lifting surfaces moving through the medium on a helical path. He took no account of the effect on each element of the velocity induced by the propeller itself.

The work of Wilbur and Orville Wright had no influence on the subsequent development of propeller theory, but it is remarkable that although they were experimenters and not theorists, they seem to have been the first to combine blade element theory and momentum theory [4]. They used momentum

theory to estimate the relative velocity and the angle of attack of blade elements and succeeded in designing quite efficient propellers.

With Prandtl's development of a lifting line theory of wings incorporating the concepts of bound and free vorticity, the way was open for a more rational theory of propeller action. Modern propeller theory is analogous to wing theory in that the propeller blade is considered to be a lifting surface about which there is a circulation associated with the bound vorticity and a vortex sheet is continuously shed from the trailing edge. In the case of a wing with spanwise load distribution for minimum energy loss, the shed vortex is a flat sheet with uniform distribution of downwash. In 1919, Betz [5] showed that the load distribution for lightly loaded propellers with minimum energy loss is such that the shed vorticity forms regular helicoidal vortex sheets moving backward undeformed behind the propeller. Prandtl found an approximate solution to the flow around helicoidal vortex sheets by likening the flow around the edges to the two-dimensional flow around a cascade of semi-infinite straight lamina. The approximation is good when the advance ratio λ is small and improves as the number of blades increases. The approximation is attractive for its simple mathematical closed form and continues to be useful.

Goldstein [6] solved the problem of the potential field and the distribution of circulation for a helicoidal vortex system for small advance ratios. The application of the solution, as he presented it, was limited to lightly loaded propellers. He presented tabulated values of the circulation distribution only for two- and four-bladed propellers.

In spite of the clearer understanding of propeller action afforded by vortex theory, the combined blade element and momentum theory continued to be refined (see [7]) and was more often employed in practical calculation even though it depended on a principle of "independence of blade elements" which eventually came to be understood to be without physical justification.

Theodorsen [8] showed that the undeforming helicoidal sheet model of the shed vorticity need not be limited in application to lightly loaded propellers. By directing attention to the vortex system far behind the propeller rather than at the propeller, the light loading limitation can be removed. Goldstein's solution for the field of a helicoidal vortex sheet when applied to the trailing vorticity far from the

propeller remains valid without regard to conditions at the propeller. Building on this important realization, Theodorsen proceeded to refine and elaborate the theory of propellers with ideal load distribution. The development is founded on analysis of helicoidal trailing vortex sheets and the thrust and torque implied by their form and displacement velocity. Conditions at the propeller and its required geometry are then developed as a consequence of the dynamics of the shed vorticity.

Accurate tabulated values of a function related to the Goldstein function and covering a wide range of parameters became available with an extensive mathematical and computational effort by Tiberly and Wrench Jr [9].

Larrabee [10] presented a practical design procedure for propellers which has the virtue of convenience, but is limited by implicit light loading assumptions and fails to take advantage of the more general concepts introduced by Theodorsen.

2. Basic principles

2.1. Kinematics and basic forces on the screw propeller

The propeller is a propulsion machine consisting of rotating lifting surfaces disposed radially about a shaft that is aligned approximately with the direction of motion. In consequence of its motion, the blade is subject to several components of relative velocity of the fluid: the axial velocity due to the velocity V through the fluid, the rotational velocity Ωr_0 and, in addition, it is subject to the induced velocity due to the disturbance of the fluid by the propeller itself.

There may also be interference due to the presence of a nacelle or a fuselage. If the propeller is behind the body it is propelling there are also wake velocity components, the most important of which is axial, but rotational and radial components may also exist. For present purposes no wake is considered.

The induced velocity may be considered to be the resultant velocity at a point due to the entire system of bound and free vorticity. For the present we assume that the blades are relatively narrow and are disposed along equally spaced radial lines. It can be seen from considerations of symmetry that equally spaced radial vortex lines of equal strength induce no net velocity on any one of the lines. Conse-

quently, these conditions assure that the effect on each blade due to bound vorticity on the other blades can be ignored and only trailing vorticity contributes to the resultant velocity at a blade.

If the blades are of sufficiently small chord, it also follows that the induced velocity does not vary significantly along the chord and the forces on the blade section are the same as would occur in a uniform velocity field. The lift on the blade element is then related to angle of attack with respect to the local relative velocity as in two-dimensional airfoil theory. This is the lifting line assumption which will be basic to this treatment. Aircraft propellers almost always may be considered to be adequately represented by the lifting line assumption. However, if the blade is relatively wide, the variation of induced velocity along the chord must be accounted for. In such cases a vortex lattice or other lifting surface representation is most appropriate. Marine propellers usually must be treated by some such scheme.

Within these limitations, the velocity at a blade section cut by a cylindrical surface is as illustrated in Fig. 1.

The fundamental expressions for the forces developed by the propeller may be expressed most conveniently by application of the Kutta–Joukowski theorem, which is conveniently applied in its vector form

$$d\mathbf{L} = \rho \mathbf{U}_0 \times \Gamma d\mathbf{r}, \quad (2.1.1)$$

where $d\mathbf{L}$ is the lift force on a blade element of radial dimension dr , $\mathbf{U}_0$ the resultant onset velocity and Γ the bound circulation. Making use of the cross product properties, we can immediately write the expressions for the two interesting components of the resultant force, the thrust and the torque. The elementary contribution of a blade element to the thrust is proportional to the product of the circulation and the component of relative velocity normal to the thrust, i.e., the rotational component

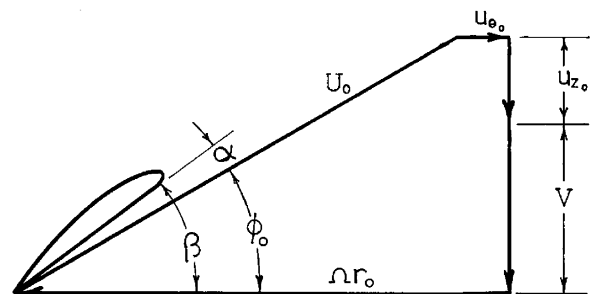


Fig. 1. Velocities at a blade element.

of U_0 . Similarly, the contribution to torque is the product of circulation and the axial component of velocity relative to the blade element:

$$dT = \rho \Gamma (\Omega r - u_{\theta_0}) dr, \quad (2.1.2)$$

$$dQ = \rho \Gamma (V + u_{z_0}) r dr, \quad (2.1.3)$$

where u_{θ_0} and u_{z_0} are velocities induced at the blade element by the free vortex systems. Eqs. (2.1.2) and (2.1.3) may be considered the fundamental relations of a lifting line theory of propellers, although they lack provision for the viscous (drag) force on the blade elements.

Integrating the elementary forces over the radius of a propeller with B blades, the thrust is

$$T = \rho B \int_{r_h}^R \Gamma (\Omega r - u_{\theta_0}) dr \quad (2.1.4)$$

and the torque is

$$Q = \rho B \int_{r_h}^R \Gamma (V + u_{z_0}) r dr. \quad (2.1.5)$$

Applying two-dimensional airfoil theory, the bound circulation on a blade element can be related to the local angle of attack of the blade element:

$$dL = c_l \frac{\rho}{2} U_0^2 c dr = a_0 (\alpha - \alpha_{L_0}) \frac{\rho}{2} U_0^2 c dr.$$

The local angle of attack $\alpha = \beta - \phi$ where β is the geometric blade angle and ϕ is the angle of the resultant relative velocity to the plane of the propeller.

Since we also have $dL = \rho U_0 \Gamma dr$,

$$\Gamma = \frac{1}{2} a_0 (\alpha - \alpha_{L_0}) U_0 c \quad (2.1.6)$$

and

$$\phi = \tan^{-1} \frac{V + u_{z_0}}{\Omega r - u_{\theta_0}}. \quad (2.1.7)$$

The complicated relation between the circulation Γ , the geometry of the blade and the induced velocities, and the difficulty of determining the induced velocity due to a trailing vortex system makes it clear that these equations by themselves are only the beginning of a complete representation of the fluid dynamics of the propeller, but they are clear fundamental relations which provide a basis for the understanding of propeller mechanics.

2.2. The thrust of a propulsive device by the momentum integral

The forces on various active or passive bodies moving in a fluid are often most profitably studied by imagining a closed control surface, S , surrounding the body, and stationary with respect to it, through which the fluid flows. The Eulerian form of the momentum theorem for a steady flow applied to such a control surface is

$$\int_S (\rho \mathbf{n} + \rho \mathbf{n} \cdot \mathbf{w} \mathbf{w}) dS - \mathbf{F} = 0. \quad (2.2.1)$$

Here, $\mathbf{n}$ is the unit outward normal to the control surface, $\mathbf{w}$ is the velocity at the surface and $\mathbf{F}$ is the resultant force acting on the fluid by the body submerged within.

Since the positive direction is downstream a positive value of $\mathbf{F}$ is a backward force on the fluid and is equal to a positive (forward) thrust on the propulsor (Fig. 2).

Consider a propulsive device located at the origin on the z -axis, which is parallel to the onset flow u_0 . Any device may be imagined so long as its effects are steady, are axially symmetric (no net radial force) and no fluid is created or destroyed by the device. A cylindrical control surface through which the fluid may freely pass is considered, its axis coinciding with the z -axis. Its ends are surfaces normal to z . Both the curved and flat surfaces are assumed to be at large distances from the propulsor.

An alternative treatment of the problem considers the curved surface S_1 to be effectively a rigid boundary through which no fluid passes. Its effects are then dismissed by asserting that it is at a sufficiently large distance from the axis that the flow is effectively unconstrained. The analysis is thus simplified, but is less illuminating and perhaps less convincing than the assumption of a control surface which does not impede the flow and serves only as a

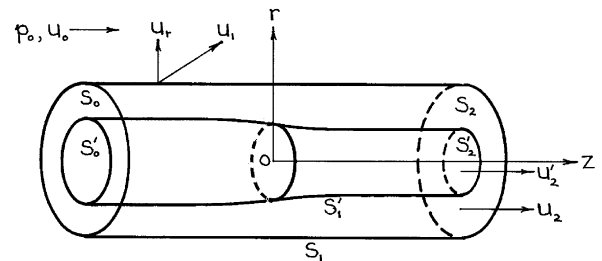


Fig. 2. Control surface for the momentum integral.

conceptual framework. This comment is generally applicable to the use of control volumes in fluid flow analysis.

The bounding planes normal to the z axis are divided into inner regions (S_0' , S_2') and outer regions (S_0 , S_2) which are separated by a stream surface S_1' bounding the fluid which is subject to direct forces by the propulsor. The integration is to be carried out over S which includes S_0 , S_0' , S_1 , S_2 and S_2' .

Considering the curved surface S_1 , it is evident from symmetry that

$$\int_{S_1} p \mathbf{n} dS_1 = 0. \quad (2.2.2)$$

Also, since $\mathbf{n} \cdot \mathbf{w} = u_r$ and $\mathbf{w} = \mathbf{u}_1$,

$$\int_{S_1} \rho \mathbf{n} \cdot \mathbf{w} dS_1 = \int_{S_1} \rho u_r \mathbf{u}_1 dS_1.$$

But $\mathbf{u}_1 = \mathbf{u}_0$ plus perturbations which become second-order quantities when multiplied by u_r . Therefore,

$$\int_{S_1} \rho \mathbf{n} \cdot \mathbf{w} dS_1 = \rho \mathbf{u}_0 \int_{S_1} u_r dS_1. \quad (2.2.3)$$

In considering velocities at the planes ahead and behind, we observe that the stream surface S_1' and the flow outside of it will remain entirely unchanged if the propulsive device is replaced by an appropriate distribution of sinks in the neighborhood of the origin. At the large distances of the control volume, the resulting contributions to the velocity will be proportional to R^{-2} where R is the distance from the origin. They become second-order perturbations which vanish in the limit when the control volume is increased in size without limit. Therefore, on S_0 , S_0' and S_2 the velocity is u_0 and the pressure is p_0 . Now, over the total area of the bounding planes of the control volume the pressure integral becomes

$$-\int_{S_0} p_0 dS_0 - \int_{S_0'} p_0 dS_0' + \int_{S_2} p_0 dS_2 + \int_{S_2'} p_2 dS_2'$$

the signs being in accord with the directions of the outward normal unit vector $\mathbf{n}$. Since $S_0 + S_0' = S_2 + S_2'$, it follows that this sum of terms is equal to

$$\int_{S_2'} (p_2 - p_0) dS_2', \quad (2.2.4)$$

which is all that remains of the pressure integral terms.

If there is no swirl behind the propeller, as in the case of the simple impulse disc, or as may be approximately true for a counter-rotating propeller, all streamlines at S_2' are straight and parallel to the axis. It is evident then that the pressure across S_2' cannot differ from p_0 . In such a case the pressure term (2.2.4) will disappear.

The momentum integral over S_1 , Eq. (2.2.3), may be evaluated by consideration of the continuity condition for the fluid within the portion of the control volume external to the slipstream surface S_1' :

$$u_0 S_0 - \int_{S_1} u_r dS_1 - u_2 S_2 = 0$$

and since, over S_2 , $u_2 = u_0$, we have for the integral on S_1 ,

$$\int_{S_1} u_r dS_1 = u_0 (S_0 - S_2).$$

Consequently, the momentum integral on S_1 , Eq. (2.2.3), becomes

$$\rho \mathbf{u}_0 u_0 (S_0 - S_2). \quad (2.2.5)$$

The momentum integrals for S_0 and S_2 are

$$-\rho u_0 \mathbf{u}_0 S + \rho u_0 \mathbf{u}_0 S_2.$$

Summing the momentum integrals on S_0 , S_1 and S_2 , we find no net contribution and we are left with the integrals on S_0' and S_1' which are

$$-\rho u_0 \mathbf{u}_0 S_0' + \int_{S_2'} \rho u_2' \mathbf{u}_2' dS_2'. \quad (2.2.6)$$

By continuity within the slipstream

$$u_0 S_0' = \int_{S_2'} u_2' dS_2',$$

hence, the total momentum integral is

$$\int_{S_2'} \rho u_2' (u_2' - u_0) dS_2'. \quad (2.2.7)$$

Adding (2.2.4) and (2.2.7), the thrust is, from Eq. (2.2.1),

$$F = \int_{S_2} [(p_2 - p_0) + \rho u_2 (u_2 - u_0)] dS_2. \quad (2.2.8)$$

The primes have been dropped since the integrand is zero outside of the slipstream and no distinction between inside and outside is necessary, S_2 being the entire plane.

If the propeller is a simple impulse disc behind which there is no swirl and where u_2 is constant, the thrust is just the mass flow rate times the increase in

velocity in the ultimate slipstream:

$$F = \rho u'_2 S'_2 (u'_2 - u_0). \quad (2.2.9)$$

The simple forms of (2.2.8) and (2.2.9) should not tempt one to apply the momentum relation to individual stream tubes without due caution. Erroneous conclusions resulting from this practice are not infrequent. Eqs. (2.2.1) and (2.2.8) are integral relations. The corresponding differential relations are generally not valid without careful consideration of a closed control surface.

3. The trailing vortex system

3.1. The condition for maximum efficiency with a given thrust

There have been many attempts to determine the condition for maximum efficiency of a propeller having a certain required thrust, beginning with the work of Betz [5] in which he considered a lightly loaded propeller. Subsequent work has sought to remove this limitation. Many of these treatments include questionable tacit assumptions, and other ill defined difficulties. The following treatment, by concentrating on the necessary conditions on the trailing vortex system as suggested by Theodorsen, seems to dispel the obscurities. Computation of the circulation on the propeller blades due to the necessary distribution of vorticity in the trailing system is then a separate problem.

This, like nearly all modern treatments of propeller theory, presumes that a trailing vortex sheet, after an initial deformation, persists for at least a moderate distance behind the propeller. However, a vortex sheet with a free edge is really a transient condition. The helicoidal vortex sheets behind a propeller will soon roll up into a set of helical vortex filaments and a central vortex filament of opposite sense on the axis. The reason for this process and its effect on conditions at the propeller will be discussed in the following section.

The quasi-steady vortex sheets are uniquely related to the induced velocities and the distribution of bound vorticity on the propeller blades. Consequently, the optimum load distribution can, in principle, be established by consideration of the helicoidal vortex sheets, even though they must eventually roll up into concentrated helical vortices and a vortex of opposite sense on the axis.

The thrust and torque of a propeller are uniquely related to the conditions some distance behind the

propeller. The effect of adding an increment of load at the propeller will be exactly the same in the ultimate wake as the addition of a corresponding increment some distance behind the propeller on the corresponding vortex sheet. To avoid the complications of the rapidly changing flow near the propeller, we consider a simple variational approach to the optimum loading by applying a load increment downstream on the trailing vortex sheets.

Consider a system of regular helicoidal vortex sheets of constant pitch. At a fixed distance from the propeller imagine radial lines, one in each of the sheets, which remain embedded in the vortex sheets. In order to remain on the sheets, the lines must rotate at angular velocity Ω . They experience a relative axial velocity $V + u_z(r)$ and tangential velocity $\Omega r - u_\theta(r)$. Now let each radial line have a bound vorticity $\varepsilon \Delta\Gamma(r)$ where ε is a parameter and $\Delta\Gamma(r)$ is a continuous function taking on both positive and negative values in the interval $0 < r < R$. The thrust and torque of an element of the bound vortex are, by the Kutta–Joukowski law, Eqs. (2.1.2) and (2.1.3),

$$dT = \rho[\Omega r - u_\theta(r)]\varepsilon\Delta\Gamma(r)dr \quad (3.1.1)$$

and

$$dQ = \rho[V + u_z(r)]\varepsilon\Delta\Gamma(r)rdr. \quad (3.1.2)$$

The parameter ε is made sufficiently small that changes of $u_\theta(r)$ and $u_z(r)$ caused by trailing vortices shed by the bound vorticity $\varepsilon\Delta\Gamma(r)$ are of second order and may be neglected. Now let $\Delta\Gamma(r)$ be any continuous function such that there is no net change in the thrust of the propeller, only a radial redistribution of thrust, i.e., the variation of the thrust must vanish:

$$\delta T = \varepsilon\rho \int_0^R [\Omega r - u_\theta(r)]\Delta\Gamma(r)dr \equiv 0. \quad (3.1.3)$$

If the vortex system emanates from a propeller with optimum radial distribution of load, the variation of torque must vanish simultaneously, otherwise the torque could be reduced and efficiency improved by a redistribution of the load. The variation of the torque with respect to the parameter ε is

$$\delta Q = \varepsilon\rho \int_0^R [V + u_z(r)]\Delta\Gamma(r)rdr. \quad (3.1.4)$$

Rather than seek a formal variational solution for the form of the trailing vortex system which satisfies Eqs. (3.1.3) and (3.1.4), it is easy to surmise the configuration and show that it is indeed a solution.

Assume that the trailing vortex system is of the form of a right helicoid in uniform axial translation. Then the velocities on the sheet must obey the following relation:

$$r \tan \phi = r[V + u_z(r)]/[\Omega r - u_\theta(r)] = \Lambda, \text{ a constant.} \quad (3.1.5)$$

In that case (3.1.4) may be written

$$\begin{aligned} \delta Q &= \varepsilon \rho \Lambda \int_0^R [\Omega r - u_\theta(r)] \Delta \Gamma(r) dr \\ &= \Lambda dT. \end{aligned}$$

Since the variation of thrust is identically zero,

$$\delta Q = \Lambda dT = 0.$$

The torque is found to be stationary when the helicoidal pitch condition Eq. (3.1.5) is imposed, independent of the specific form of the function $\Delta \Gamma(r)$, i.e. under the imposed conditions there is no function $\Delta \Gamma(r)$ which will reduce the torque. Consequently, the assumption Eq. (3.1.5) is shown to be the condition for optimum loading. Since the condition for maximum efficiency is $r \tan \phi = \Lambda$, a constant, the optimum distribution of circulation is realized when the trailing vortex sheets are helicoids of uniform pitch which move backward undeformed.

No assumption has been made as to whether the basic loading is light or heavy, i.e., no restriction is imposed on the magnitude of w/V , the relative velocity of the helicoidal vortex sheets or of the magnitude of the initial radial displacement of the vortex trails before they become a set of substantially undeforming helicoidal sheets. By considering an increment of circulation in the free vortex system, the effect of the incremental load is obtained by the Kutta–Joukowski theorem without any complication due to pressure effects which are frequently cited as complicating factors when the loading is heavy.

It is established that *efficiency is a maximum without restriction of loading when the pitch of trailing vortices is constant and each trailing vortex sheet translates backward as an undeforming regular helicoidal surface.*

It remains to evaluate the distribution of circulation of the vortex sheet which corresponds to the prescribed configuration and to relate this to the configuration of the propeller itself.

By concentrating attention entirely on the trailing vortex system and considering an incremental load thereon, doubts that have been raised regard-

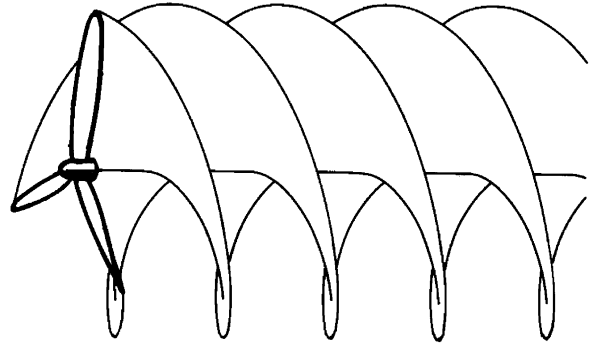


Fig. 3. Propeller with trailing helicoidal vortex sheets.

ing the validity of various optimizing techniques are largely obviated. The result, while clear, leaves difficulties regarding application to the geometry of the propeller itself as distinct from the configuration of the trailing vorticity of the optimum propeller.

It may be observed that the marginal efficiency associated with a small *increment* of circulation $d\Gamma$ at radius r is

$$\begin{aligned} \eta_m &= \frac{V \delta T}{\Omega \delta Q} = \frac{V(\Omega r - w \cos \phi \sin \phi)}{\Omega r(V + w \cos^2 \phi)} \\ &= (V/\Omega r) \cot \phi = V/(V + w). \end{aligned} \quad (3.1.6)$$

This is a constant for the helicoidal vortex sheet. It has sometimes been said that an optimum propeller must have constant efficiency along the blade, i.e. at all radii, but this is plainly not true. It is the marginal efficiency associated with the last increment of load at an element which must be constant with respect to radius (Fig. 3).

3.2. Kinematics of the helicoidal vortex sheet

For a point on each of B undeforming surfaces streaming backward from B equally spaced propeller blades

$$\theta = \Omega t + 2\pi(n-1)/B, \quad n = 1, 2, \dots, B$$

and

$$z = (V + w)t,$$

where w is the backward velocity of the sheet with respect to the surrounding fluid. Eliminating t , the equation of the sheet is

$$\theta - \frac{\Omega}{V + w} z - 2\pi(n-1)/B = 0. \quad (3.2.1)$$

The angular pitch of the sheet is

$$\tan \phi = \frac{dz}{r d\theta} = (V + w)/\Omega r = \lambda_2/x_1 \quad (3.2.2)$$

and the linear pitch is

$$P = 2\pi r \tan \phi = 2\pi(V + w)/\Omega = 2\pi R_1 \lambda_2. \quad (3.2.3)$$

The velocity normal to the sheet at any point on its surface is the velocity with which it is convected freely by the established fluid motion and is simply the normal component, $w \cos \phi$, of the displacement velocity of the sheet. It is evident that the axial and tangential components of the convective velocity of the sheet are

$$u_{z_1} = w \cos^2 \phi \quad u_{\theta_1} = w \cos \phi \sin \phi \quad (3.2.4)$$

and

$$\tan \phi = (V + u_{z_1})/(\Omega r - u_{\theta_1}). \quad (3.2.5)$$

From Eq. (3.2.2) it follows that

$$u_{z_1} = w/(1 + \lambda_2^2/x_1^2) \quad (3.2.6)$$

and

$$u_{\theta_1} = w(\lambda_2/x_1)/(1 + \lambda_2^2/x_1^2). \quad (3.2.7)$$

It has been pointed out [10] that the helicoidal vortex sheets are locally convected normal to themselves and therefore they have a rotational velocity as well as translation along the axis. Whether a helicoid rotates around its axis, translates along the axis, or both simultaneously, it will look exactly the same. The vortex sheet does not contain fluid particles but is a sheet of velocity discontinuity in the fluid. Consequently, the distinction between rotation and translation is meaningless. The flow around the helicoids and the momentum transport is in any case exactly the same as for axially translating helicoidal surfaces (Fig. 4).

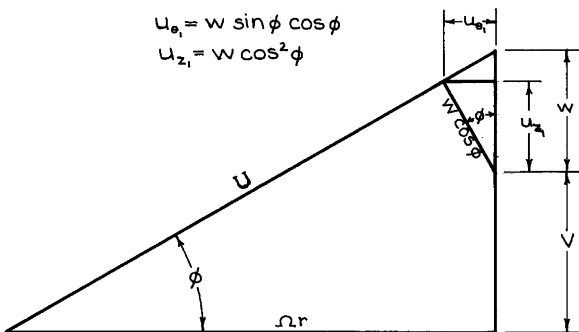


Fig. 4. Components of velocity at the helicoidal vortex sheet.

3.3. The dynamics of trailing vortex sheets

A consequence of Helmholtz's vortex theorems [11] is that a vortex cannot end in a fluid. Consequently, when the bound vorticity on a lifting surface varies in magnitude along the span, a free vortex filament must emanate from the trailing edge with magnitude equal to the change of bound vorticity. The derivative of the strength of the free vortex sheet in the spanwise direction must be equal to the negative of the derivative of the strength of the bound vorticity in the spanwise direction. Letting Γ_B be the magnitude of the bound vorticity and Γ_F the free vorticity, this may be expressed as $d\Gamma_F/dr = -d\Gamma_B/dr$. The shed vortex filaments constitute the trailing vortex sheet that must exist wherever bound vorticity is not constant along the span.

The vortex sheet may be thought of as drifting with the fluid. There can be no forces on it, no discontinuity of pressure, and no discontinuity of normal velocity, only a discontinuity of tangential velocity the magnitude of which is the vortex strength of the sheet.

A lifting surface and its bound vorticity are represented in Fig. 5 together with the trailing vortex sheet. Consider a point p on the lower surface of the vortex sheet and an adjacent point p' on the upper surface. Connect the two points by an arbitrary path s which passes around the edge of the sheet from the lower side of the sheet to the upper, enclosing all of the vorticity between p and the edge of the sheet. The integral

$$\int_p^{p'} \mathbf{u} \cdot d\mathbf{s} = \Delta\phi$$

is the difference in potential between points p and p' . If now we penetrate the sheet, letting p and p' come

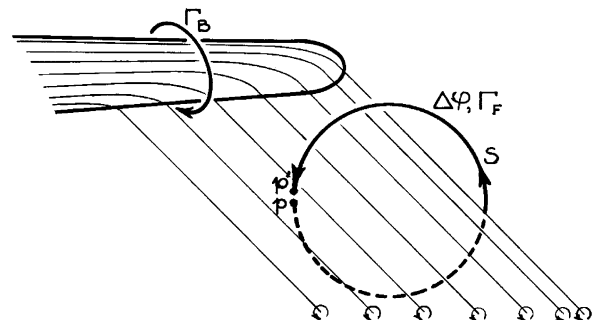


Fig. 5. Bound and free vorticity.

together, the integral around the closed path is just the total vortex strength between pp' and the free edge since the path of integration encloses all of the vortex lines in this part of the vortex sheet:

$$\oint \mathbf{u} \cdot d\mathbf{s} = \Gamma_F.$$

Since the path of integration is unchanged, the foregoing integrals are of identical magnitude and $\Delta\phi = \Gamma_F$. *The potential difference across a vortex sheet at any point on the sheet is equal to the total circulation between the point and the edge of the sheet.*

Now consider a path of integration around the lifting surface where the circulation is Γ_B . Locate pp' so that a line may be drawn from the trailing edge where the circulation is Γ_B to pp' without crossing a vortex filament, i.e. a line lying along the vortex lines of the trailing sheet. It is then evident from the diagram that the integration around the lifting surface encloses the same vorticity as the path s around the trailing vortex sheet and $\Gamma_F = \Gamma_B$. Therefore, $\Gamma_B = \Delta\phi$. *The bound circulation on a lifting surface is equal to the potential difference across the trailing vortex sheet at a corresponding point.*

The behavior of vortex sheets as they exist behind wings has been extensively studied. The vortex sheet shed by an elliptically loaded wing is initially a flat sheet of width equal to the wing span and extending downstream without deformation. In the Trefftz plane, a plane normal to the relative wind far downstream, the vortex sheet is a simple slit, a straight line segment moving normal to itself in a two-dimensional flow. It provides a much more tractable subject for analysis than the helicoidal sheets behind a propeller. Consequently, the difficulties arising around the question of the existence and the stability of vortex sheets are discussed here in the context of the vortex sheet as a plane lamina in the wake of a lifting surface.

3.4. The edge force and the rolling up of a vortex sheet

Although the pressures on either side of the sheet are equal, suggesting that it may translate freely without deformation, there is a serious difficulty at the edge of the sheet where there is a singularity in the velocity field. It will be shown that there is an edge force on a plane lamina moving normal to a stream. The simplest way to demonstrate the

existence of the edge force is to consider an ellipse immersed in a uniform stream parallel to its minor axis. The reduced pressure on the halves of the ellipse tends to pull it apart. Letting the minor axis of the ellipse $b \rightarrow 0$, it degenerates to a plane lamina with a tensile force at its edge.

Consider an elliptic cylinder $(x/a)^2 + (y/b)^2 = 1$ where a is the major semi-axis. The ellipse is immersed in a uniform stream of velocity U in the negative y direction. The tangential velocity at the surface of the ellipse is [12, p. 199] or [13, p. 181]:

$$u/U = \frac{(a+b)(x/a)}{[b^2 + (c^2/b^2)y^2]^{1/2}} \quad \text{where } c^2 = a^2 - b^2.$$

Letting $\eta = y/b$ and $\beta = b/a$,

$$u/U = \frac{(1+\beta)(1-\eta^2)^{1/2}}{[\beta^2 + (1-\beta^2)\eta^2]^{1/2}}.$$

The surface pressure is

$$p/\frac{1}{2}\rho U^2 = 1 - (u/U)^2.$$

The force tending to pull the ellipse apart, i.e. the suction on either half tending to pull it out normal to the stream in the direction of the major semi-axis is

$$F_x = - \int_{-b}^{+b} p \, dy = -2 \int_0^b p \, dy$$

from which we find

$$F_x/\frac{1}{2}\rho U^2 = \frac{-2a}{1-\beta} \left\{ 2\beta - \sqrt{\frac{1+\beta}{1-\beta}} \tan^{-1} \left(\frac{\sqrt{1-\beta^2}}{\beta} \right) \right\}.$$

Flattening the ellipse to a plane lamina by letting $\beta \rightarrow 0$, the force tending to stretch the lamina in the x direction is found to be

$$F_x = \frac{1}{2}\pi a \rho U^2. \quad (3.4.1)$$

Consequently, the plane vortex sheet model of the field implies a very substantial force at the edge where there is a singularity in the velocity field. Since there is no rigid body on which such a point force can act and the vortex sheet cannot support tension, the postulated model of the flow is dynamically inconsistent and cannot exist except as an idealized transient. In reality the vortex sheet will have some finite thickness initially equal to the combined thickness of the boundary layers shed from the upper and lower surfaces of the wing (or propeller blade). Such a vortex layer will not have the singularity at its edge which appears with the

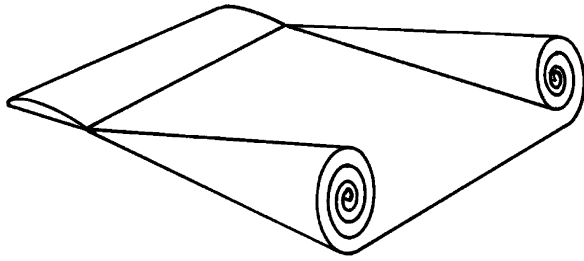


Fig. 6. Rolling up of the vortex sheet behind a wing.

idealized sheet of zero thickness, but there will still be a high velocity and attendant low pressure tending to pull the vortex layer out at its edge.

It is found that the stretching of the vortex sheet caused by the low pressure at its edge causes, bit by bit, entrainment of the vortex sheet by the high velocity fluid. This results in the rolling up of the sheet around a streamwise core. The rolling up of the vortex sheet begins as an infinitesimal vortex filament at its edge and proceeds as a growing vortex which continuously feeds on the vortex sheet, pulling it out in a spiral around the initial vortex. Eventually, the vortex sheet is completely absorbed in a vortex of some non-zero diameter (not a vortex filament) whose axis remains near the position of the edge of the disappearing sheet (Fig. 6).

While it is true that a vortex sheet is unstable away from any edge, the rolling up of the edge of a vortex sheet, properly speaking, is not due to instability as has often been said. Instability is the condition of a system in equilibrium responding to a small disturbance by an ever-growing change of the dynamics of the system. A vortex sheet with an edge is not a system in equilibrium, but a transient distribution of vorticity that cannot be in equilibrium [11, pp. 97–101]. Rolling up is the necessary consequence of the singularity or the extreme pressure gradients that exist near the edge.

3.5. The helicoidal vortex sheet

Suppose that, after an initial distortion, the vortex sheets shed from the trailing edges of the propeller blades form a set of interleaved helicoidal sheets which translate uniformly downstream parallel to the axis without further deformation as if they were rigid surfaces. The change in radial velocity across the sheet is the vortex strength of the sheet and everywhere has the magnitude required for it to be in equilibrium. The helicoidal vortex sheets are floating freely in an irrotational

field with equal velocity on either side of the sheet, hence equal pressure. Since there is no pressure discontinuity across the sheets, it may be hypothesized that the sheets move axially backward without deformation. The system of helicoidal vortex sheets moving backward without deformation is a mathematical model which provides a means of connecting the induced velocity at the propeller with the propeller loading. Most importantly, under certain assumptions it has been shown to be the slipstream condition for maximum efficiency for a given required thrust. Consequently, it dictates the radial load distribution on the propeller blades for best efficiency. For these reasons, it is the essential framework for a propeller design system.

Vortex sheets are considered to be of vanishing thickness, simple surfaces of velocity discontinuity. All of the fluid in the slipstream is contained between the vortex sheets and is therefore everywhere irrotational even as the distance between sheets becomes vanishingly small. It is evident that in the limit $B \rightarrow \infty$ this does not represent a physically meaningful flow. In a real fluid the sheets always have some thickness and in the limit the fluid must be filled with vorticity. The vortex sheet treatment is only valid where the distance between the sheets is at least comparable with the thickness of the sheets.

Passing from the case of the plane vortex sheet behind a wing to the case of the postulated helicoidal sheets behind a propeller, the outer parts of the sheets are absorbed into a set of helical vortices equal in number to the number of interleaved sheets and the inner parts are absorbed in a single vortex of opposite sense lying on the axis. Freely moving helicoidal vortex sheets in the slipstream of a propeller would seem to be an unrealistic hypothesis in view of the necessity of an edge force with nothing on which to act. However, they can and do exist in the modified model of helicoidal sheets which are more or less gradually absorbed into a set of helical vortices.

Several arguments may be put forth to justify the helicoidal vortex sheets as adequate representations of the trailing vortex system for the purpose of relating the loading of the propeller to the velocities induced by the trailing vortices at the propeller blades. First consider the following two principles [11, pp. 99–103, 14, pp. 517–520]:

- In the evolution of a free vortex system in the absence of external forces acting on the fluid, hydrodynamic impulse is conserved.

- If in an unbounded fluid at rest at infinity there is a vortex system having a certain impulse, replacement of the vortex system by another of the same impulse may result in a very different distribution of velocity in the neighborhood of the vortex, but the velocity fields will be identical at large distances.

From these two principles it is inferred that the velocities induced at the propeller by downstream portions of the fully rolled-up helical vortex system are the same as would be induced by undeforming helicoidal vortex sheets. Immediately behind the propeller there are helicoidal vortex sheets. It is only the part of the vortex system in an intermediate region where the sheets are rolling up that there may be some doubt of the accuracy of the helicoidal sheet model as contributor to the velocity induced at the propeller.

Consequently, it is justifiable to use the mathematical model of helicoidal vortex sheets translating backward without deformation as the condition for a propeller with ideal load distribution. This is to be understood as a special case since for arbitrary radial distribution of circulation the axial induced velocity of the trailing vortices will not be uniform and the vortex sheets will have a continuously changing form. The vortex system of heavily loaded propellers may, in some circumstances, roll up in quite strange and unexpected ways.

3.6. The Goldstein circulation function

Having established that the shed vortex system behind a propeller with ideal load distribution may be represented as a regular helicoidal vortex sheet moving uniformly backward in the fluid, it is required to determine the distribution of vorticity on such a vortex sheet and deduce from this the bound circulation on the propeller. This necessitates the determination of the potential function ϕ which describes the flow in the surrounding fluid. The partial differential equation that must be satisfied by ϕ is

$$\nabla^2 \phi = 0$$

and the boundary condition is that the normal velocity everywhere on the surface defined by Eq. (3.2.1) is

$$\partial \phi / \partial n = w \cos \tau.$$

(Here τ is the pitch angle of the helicoidal sheet, used here to avoid confusion with the potential.)

The determination of the potential function ϕ is an uncommonly difficult problem the details of which we need not consider here.

Goldstein [6] found a solution to the general potential problem. He considered a lightly loaded propeller and succeeded in calculating the potential function for two bladed and four-bladed propellers over a range of advance ratios. However, such are the difficulties of computation, even after the way to a solution was found, that Theodorsen in his intensive study of propellers at the National Advisory Committee for Aeronautics, the results of which were published in 1944, resorted to the use of a rheoelectrical analog to evaluate the circulation function. (His work was finally published in his book [8].)

Theodorsen made an important contribution to the problem by pointing out that Goldstein's limitation of his analysis to the case of lightly loaded propellers is unnecessary if it is realized that the circulation function is dependent only on the configuration of the helicoidal sheets at a distance behind the propeller. It is not necessary that the pitch there be the same as at the propeller, as would be the case when the loading is light. The circulation $\Gamma(r_1)$ is the strength of the vortex sheet downstream where it is moving like a rigid helicoid at some axial velocity w with respect to the surrounding fluid. It becomes a separate problem to trace the vortex filaments back to the propeller to find the corresponding point where $\Gamma(r_0) = \Gamma(r_1)$, thus defining the bound circulation and the loading on the propeller blade.

The Goldstein circulation function as originally defined by him and as used in the extensive studies of Theodorsen is the circulation $\Gamma(r_1)$ on the trailing vortex sheet expressed as a dimensionless factor

$$G(r_1) = \Gamma(r_1) / hw, \quad (3.6.1)$$

where h is the axial distance between adjacent turns of the helicoidal sheets and w the backward velocity of the vortex system with respect to the surrounding fluid. $G(r)$ is, of course, dependent on the geometry of the vortex system as defined by λ , the pitch of the helicoid, and B , the number of interleaved sheets. Since $h = P/B = 2\pi(V + w)/\Omega B$, the Goldstein function may be written

$$G(r_1) = B\Gamma\Omega/2\pi w(V + w) = B\Gamma/2\pi R_1 w \lambda_2 \quad (3.6.2)$$

Goldstein, assuming light loading, wrote V where we have $(V + w)$.

Ribner and Foster in a more recent study [15] generated Goldstein functions by representing the trailing sheets by sets of discrete helical vortex filaments with strengths adjusted to produce the uniform backward translation. This is a more physical representation of the problem, but is less accurate and they covered a limited range of variables. That Theodorsen resorted to a rheoelectric analog and Ribner and Foster employed a numerical solution to a discretized representation attests to the difficulty and complexity of an analytical solution for the potential field of a set of interleaved helicoidal sheets.

Accurate tabulated values of $G(r)$ for a wide range of parameters became available with an extensive mathematical and computational effort by Tibery and Wrench Jr [9] which culminated in tables giving accurate values for all numbers of blades from two to ten and λ_2 from 1/12 to 4.0. (This work was apparently unknown to Ribner and Foster.) These tables actually define the function in a different manner. It is taken as the ratio of the circulation $\Gamma(r)$ to the circulation which would obtain if there were an infinite number of sheets (infinite number of blades). Designating their tabulated function as $K(r)$

$$K(r_1) = B\Gamma/\Gamma_\infty = B\Gamma/2\pi r_1 u_\theta \quad (3.6.3)$$

recalling Eq. (3.2.7)

$$u_\theta = w(\lambda_2/x_1)/(1 + \lambda_2^2/x_1^2).$$

Hence,

$$K(r_1) = G(r_1)(1 + \lambda_2^2/x_1^2). \quad (3.6.4)$$

Consequently, the values tabulated in Tibery and Wrench must be divided by $(1 + \lambda_2^2/x_1^2)$ to obtain the function $G(x)$ as defined by Goldstein. Actually, there are slight advantages for either form, but the form used by Goldstein and Theodorsen is more graphic in that $G(r)$ or $G(x)$ shows the actual shape of the circulation distribution with respect to radius.

A table of the Goldstein circulation function $G(x)$ for blade numbers from two through six obtained by conversion of the tables of Tibery and Wrench is presented as Table 1.

3.7. Prandtl's approximate solution for the circulation function

Prandtl [5,7] proposed an approximate solution for the potential flow around a set of translating helicoidal surfaces by likening the flow around the edges to the flow around a two-dimensional set of equally spaced semi-infinite lamina (Fig. 7). This solution has often been presented as a “tip loss correction” for the thrust as a consequence of a finite number of blades, but it affords a good estimate of the circulation distribution for the outer parts of the propeller blade, especially at lower advance ratios and larger numbers of blades. Goldstein in the paper in which he deduced the circulation distribution for the helicoidal vortex sheets compares Prandtl's approximate solution with his own. Prandtl's treatment continues to be of interest because it provides a simple closed form expression for the circulation where more exact solutions are only available as tabulated functions based on some formidable mathematics.

Since the approximate representation of the vortex system is two dimensional, it must be applied as a modification to a simple three-dimensional representation where the fluid entrained by the helicoidal vortex sheets is entirely carried along without loss of velocity between the sheets. The tangential component of velocity is then, by Eq. (3.2.7),

$$u_\theta = w(\lambda_2/x)/(1 + \lambda_2^2/x^2).$$

The circulation is then

$$\Gamma = 2\pi r u_\theta / B = \frac{2\pi R w}{B} \frac{\lambda_2}{1 + \lambda_2^2/x^2}$$

and

$$\frac{B\Gamma\Omega}{2\pi(V+w)w} = \frac{x^2}{x^2 + \lambda_2^2}. \quad (3.7.1)$$

Turning to the two-dimensional model, the complex potential for the flow normal to the set of semi-infinite straight lamina is

$$W = \varphi + i\psi = -v'(s/\pi) \cos^{-1} e^{\pi z/s}, \quad (3.7.2)$$

where v' is the velocity of the external stream and s is the spacing of the lamina.

At any point P on one of the lamina, the difference in potential between the two sides is

$$\Delta\varphi_P = 2v'(s/\pi) \cos^{-1} e^{-\pi a/s}, \quad (3.7.3)$$

where a is the distance from the edge of the lamina. The jump in the potential is equal to the circulation around the lamina between the point P and its edge.

Table 1

The Goldstein function $G(x, \lambda_2)$ $B = 2$

x	0.2	0.3	0.4	0.5	0.6	0.7
$1/\lambda_2$						
0.25	.00385	.00562	.00719	.00847	.00936	.00972
0.50	.01500	.02181	.02779	.03258	.03581	.03691
0.75	.03248	.04703	.05954	.06929	.07548	.07706
1.00	.05517	.07941	.09976	.11506	.12410	.12538
1.25	.08188	.11706	.14583	.16660	.17789	.17794
1.50	.11152	.15826	.19541	.22114	.23394	.23194
1.75	.14308	.20148	.24658	.27656	.29011	.28547
2.00	.17572	.24547	.29782	.33128	.34497	.33735
2.25	.20873	.28922	.34799	.38419	.39757	.38683
2.50	.24158	.33198	.39628	.43457	.44732	.43351
2.75	.27382	.37321	.44215	.48197	.49392	.47721
3.00	.30518	.41254	.48529	.52617	.53722	.51785
3.25	.33542	.44974	.52553	.56708	.57722	.55549
3.50	.36444	.48472	.56284	.60474	.61400	.59023
3.75	.39215	.51744	.59725	.63924	.64770	.62222
4.00	.41854	.54794	.62889	.67076	.67850	.65163
4.25	.44357	.57628	.65788	.69946	.70657	.67877
4.50	.46734	.60258	.68440	.72555	.73212	.70341
4.75	.48989	.62696	.70862	.74923	.75536	.72614
5.00	.51120	.64953	.73071	.77069	.77646	.74698
5.25	.53140	.67043	.75086	.79013	.79562	.76610
5.50	.55052	.68978	.76923	.80773	.81300	.78363
5.75	.56869	.70770	.78598	.82366	.82878	.79974
6.00	.58589	.72432	.80125	.83808	.84310	.81452
7.00	.64642	.77971	.85021	.88336	.88830	.86263
8.00	.69603	.82137	.88463	.91400	.91913	.89725
9.00	.73713	.85326	.90927	.93500	.94032	.92240
10.00	.77150	.87807	.92726	.94964	.95507	.94083
11.00	.80045	.89762	.94063	.96005	.96545	.95445
12.00	.82497	.91324	.95078	.96760	.97286	.96460

The Goldstein function $G(x, \lambda_2)$ $B = 2$

x	0.8	0.85	0.90	0.925	0.950	0.975
$1/\lambda_2$						
0.25	.00929	.00864	.00754	.00673	.00564	.00405
0.50	.03502	.03244	.02818	.02512	.02104	.01510
0.75	.07236	.06668	.05764	.05124	.04283	.03074
1.00	.11649	.10679	.09186	.08148	.06798	.04879
1.25	.16376	.14944	.12800	.11332	.09440	.06775
1.50	.21172	.19250	.16433	.14526	.12087	.08674
1.75	.25889	.23471	.19984	.17646	.14670	.10526
2.00	.30437	.27535	.23399	.20645	.17153	.12307
2.25	.34766	.31402	.26652	.23501	.19518	.14006
2.50	.38851	.35055	.29727	.26207	.21760	.15617
2.75	.42683	.38488	.32625	.28757	.23878	.17141
3.00	.46262	.41704	.35347	.31158	.25874	.18579
3.25	.49596	.44710	.37903	.33416	.27755	.19938
3.50	.52697	.47518	.40301	.35539	.29529	.21223
3.75	.55575	.50139	.42551	.37538	.31203	.22438
4.00	.58248	.52587	.44666	.39422	.32785	.23589
4.25	.60730	.54874	.46655	.41199	.34283	.24683
4.50	.63035	.57014	.48528	.42880	.35703	.25724
4.75	.65178	.59018	.50297	.44472	.37054	.26716

Table 1 (continued)

The Goldstein function $G(x, \lambda_2)$ $B = 2$

x	0.8	0.85	0.90	0.925	0.950	0.975
5.00	.67173	.60898	.51969	.45983	.38342	.27666
5.25	.69031	.62663	.53552	.47419	.39570	.28575
5.50	.70762	.64324	.55055	.48789	.40746	.29449
5.75	.72379	.65889	.56483	.50096	.41872	.30290
6.00	.73890	.67365	.57843	.51346	.42954	.31100
7.00	.79037	.72517	.62703	.55863	.46904	.34087
8.00	.83048	.76705	.66816	.59756	.50370	.36751
9.00	.86218	.79808	.70357	.63173	.53466	.39168
10.00	.88748	.83054	.73442	.66209	.56267	.41391
11.00	.90785	.85487	.76155	.68933	.58825	.43454
12.00	.92432	.87548	.78557	.71393	.61178	.45384

The Goldstein function $G(x, \lambda_2)$ $B = 3$

x	0.2	0.3	0.4	0.5	0.6	0.7
$1/\lambda_2$						
0.25	.00408	.00661	.00900	.01105	.01257	.01332
0.50	.01588	.02565	.03476	.04244	.04797	.05049
0.75	.03433	.05512	.07417	.08985	.10065	.10489
1.00	.05808	.09254	.12345	.14811	.16425	.16940
1.25	.08574	.13540	.17885	.21241	.23317	.23814
1.50	.11604	.18140	.23714	.27879	.30315	.30697
1.75	.14790	.22867	.29576	.34432	.37123	.37325
2.00	.18043	.27577	.35287	.40701	.43557	.43544
2.25	.21301	.32168	.40722	.46565	.49512	.49279
2.50	.24516	.36565	.45806	.51960	.54944	.54503
2.75	.27658	.40733	.50504	.56864	.59845	.59220
3.00	.30703	.44646	.54805	.61282	.64232	.63455
3.25	.33640	.48298	.58718	.65237	.68136	.67241
3.50	.36464	.51692	.62261	.68761	.71598	.70617
3.75	.39172	.54839	.65461	.71890	.74657	.73623
4.00	.41760	.57751	.68346	.74665	.77356	.76296
4.25	.44239	.60443	.70944	.77121	.79735	.78672
4.50	.46604	.62932	.73284	.79297	.81830	.80785
4.75	.48861	.65232	.75392	.81222	.83675	.82666
5.00	.51015	.67360	.77292	.82927	.85300	.84339
5.25	.53067	.69328	.79008	.84439	.86734	.85830
5.50	.55025	.71151	.80559	.85782	.87998	.87159
5.75	.56895	.72841	.81962	.86976	.89115	.88345
6.00	.58675	.74408	.83235	.88039	.90103	.89405
7.00	.65005	.79641	.87274	.91281	.93060	.92647
8.00	.70233	.83581	.90094	.93406	.94930	.94761
9.00	.74557	.86585	.92114	.94851	.96155	.96165
10.00	.78140	.88903	.93597	.95869	.96984	.97116
11.00	.81118	.90712	.94710	.96611	.97567	.97773
12.00	.83601	.92140	.95562	.97168	.97991	.98237

The Goldstein function $G(x, \lambda_2)$ $B = 3$

x	0.8	0.85	0.9	0.925	0.950	0.975
$1/\lambda_2$						
0.25	.01295	.01213	.01065	.00954	.00804	.00583
0.50	.04866	.04539	.03970	.03550	.02987	.02165
0.75	.10006	.09283	.08077	.07205	.06048	.04377

Table 1 (continued)

The Goldstein function $G(x, \lambda_2)$ $B = 3$

x	0.8	0.85	0.9	0.925	0.950	0.975
1.00	.15992	.14762	.12780	.11373	.09527	.06886
1.25	.22276	.20473	.17652	.15681	.13113	.09468
1.50	.28501	.26106	.22441	.19907	.16628	.11998
1.75	.34456	.31485	.27008	.23937	.19978	.14411
2.00	.40032	.36523	.31290	.27717	.23124	.16679
2.25	.45179	.41184	.35264	.31232	.26054	.18796
2.50	.49890	.45466	.38930	.34484	.28773	.20763
2.75	.54176	.49383	.42306	.37486	.31290	.22593
3.00	.58062	.52960	.45411	.40259	.33625	.24296
3.25	.61581	.56225	.48270	.42824	.35794	.25885
3.50	.64764	.59207	.50908	.45202	.37815	.27373
3.75	.67646	.61933	.53346	.47412	.39703	.28769
4.00	.70256	.64431	.55606	.49472	.41106	.30086
4.25	.72623	.66725	.57708	.51399	.43139	.31332
4.50	.74773	.68835	.59666	.53208	.44712	.32515
4.75	.76729	.70782	.61498	.54910	.46200	.33642
5.00	.78512	.72580	.63214	.56515	.47615	.34718
5.25	.80139	.74246	.64827	.58034	.48961	.35749
5.50	.81626	.75790	.66346	.59475	.50247	.36739
5.75	.82987	.77226	.67779	.60844	.51477	.37693
6.00	.84235	.78564	.69134	.62148	.52655	.38613
7.00	.88294	.83081	.73888	.66807	.56939	.42005
8.00	.91222	.86559	.77792	.70747	.60666	.45030
9.00	.93362	.89271	.81042	.74131	.63957	.47772
10.00	.94941	.91403	.83774	.77067	.66897	.50283
11.00	.96113	.93090	.86086	.79632	.69542	.52603
12.00	.96989	.94430	.88050	.81884	.71937	.54758

The Goldstein function $G(x, \lambda_2)$ $B = 4$

x	0.2	0.3	0.4	0.5	0.6	0.7
$1/\lambda_2$						
0.25	.00393	.00698	.01007	.01286	.01505	.01628
0.50	.01530	.02707	.03882	.04926	.05725	.06150
0.75	.03308	.05803	.08251	.10378	.11949	.12711
1.00	.05595	.09706	.13652	.16990	.19356	.20378
1.25	.08255	.14134	.19638	.24159	.27230	.28391
1.50	.11166	.18842	.25834	.31408	.35043	.36232
1.75	.14231	.23638	.31960	.38403	.42452	.43588
2.00	.17371	.28379	.37826	.44940	.49265	.50304
2.25	.20533	.32969	.43322	.50912	.55397	.56322
2.50	.23674	.37350	.48390	.56281	.60835	.61649
2.75	.26766	.41491	.53013	.61056	.65607	.66325
3.00	.29790	.45373	.57200	.65272	.69766	.70406
3.25	.32734	.49002	.60974	.68976	.73374	.73957
3.50	.35583	.52375	.64368	.72222	.76495	.77040
3.75	.38340	.55507	.67414	.75063	.79191	.79716
4.00	.40995	.58412	.70148	.77548	.81519	.82038
4.25	.43547	.61102	.72601	.79725	.83531	.84053
4.50	.45996	.63592	.74805	.81634	.85272	.85805
4.75	.48344	.65896	.76787	.83310	.86780	.87329
5.00	.50595	.68029	.78570	.84787	.88089	.88657
5.25	.52742	.70003	.80180	.86091	.89230	.89817
5.50	.54795	.71831	.81633	.87245	.90225	.90831
5.75	.56752	.73523	.82948	.88269	.91096	.91721

Table 1 (continued)

The Goldstein function $G(x, \lambda_2)$ $B = 4$

x	0.2	0.3	0.4	0.5	0.6	0.7
6.00	.58619	.75092	.84140	.89180	.91862	.92501
7.00	.65240	.80308	.87922	.91968	.94132	.94806
8.00	.70668	.84195	.90561	.93819	.95573	.96239
9.00	.75108	.87127	.92454	.95104	.96536	.97165
10.00	.78746	.89365	.93846	.96027	.97211	.97875
11.00	.81733	.91097	.94893	.96713	.97702	.98218
12.00	.84197	.92456	.95698	.97235	.98072	.98532

The Goldstein function $G(x, \lambda_2)$ $B = 4$

x	0.8	0.85	0.9	0.925	0.950	0.975
$1/\lambda_2$						
0.25	.01607	.01515	.01339	.01204	.01018	.00742
0.50	.06023	.05655	.04976	.04464	.03768	.02745
0.75	.12322	.11509	.10075	.09015	.07592	.05519
1.00	.19555	.18176	.15837	.14138	.11882	.08624
1.25	.27009	.25005	.21706	.19345	.16232	.11768
1.50	.34233	.31603	.27365	.24361	.20421	.14796
1.75	.40983	.37766	.32655	.29053	.24343	.17634
2.00	.47148	.43409	.37516	.33375	.27963	.20261
2.25	.52699	.48515	.41942	.37323	.31282	.22678
2.50	.57654	.53107	.45955	.40919	.34320	.24899
2.75	.62057	.57225	.49593	.44195	.37102	.26946
3.00	.65959	.60914	.52893	.47186	.39659	.28837
3.25	.69414	.64224	.55892	.49926	.42016	.30593
3.50	.72475	.67198	.58630	.52444	.44199	.32231
3.75	.75190	.69875	.61136	.54768	.46229	.33766
4.00	.77601	.72292	.63437	.56921	.48126	.35211
4.25	.79747	.74481	.65559	.58923	.49906	.36578
4.50	.81658	.76466	.67521	.60791	.51580	.37874
4.75	.83366	.78272	.69340	.62541	.53161	.39108
5.00	.84894	.79918	.71031	.64183	.54660	.40287
5.25	.86263	.81422	.72609	.65729	.56083	.41416
5.50	.87492	.82799	.74082	.67186	.57438	.42500
5.75	.88597	.84061	.75460	.68564	.58731	.43543
6.00	.89592	.85219	.76753	.69870	.59966	.44548
7.00	.92690	.89001	.81192	.74467	.64423	.48252
8.00	.94772	.91745	.84704	.78258	.68244	.51539
9.00	.96192	.93758	.87513	.81425	.71566	.54500
10.00	.97173	.95247	.89778	.84092	.74483	.57196
11.00	.97858	.96354	.91612	.86352	.77059	.59667
12.00	.98343	.97181	.93106	.88276	.79348	.61948

The Goldstein function $G(x, \lambda_2)$ $B = 5$

x	0.2	0.3	0.4	0.5	0.6	0.7
$1/\lambda_2$						
0.25	.00367	.00703	.01066	.01410	.01694	.01871
0.50	.01433	.02724	.04099	.05388	.06429	.07047
0.75	.03106	.05834	.08687	.11302	.13350	.14491
1.00	.05266	.09745	.14315	.18393	.21481	.23071
1.25	.07792	.14170	.20492	.25974	.29980	.31883
1.50	.10578	.18864	.26825	.33520	.38251	.40330
1.75	.13535	.23640	.33027	.40684	.45933	.48082

Table 1 (continued)

2.00	.16593	.28366	.38919	.47273	.52847	.54996
2.25	.19701	.32951	.44401	.53203	.58941	.61049
2.50	.22816	.37336	.49429	.58463	.64237	.66283
2.75	.25912	.41491	.53997	.63087	.68389	.70776
3.00	.28964	.45400	.58124	.67128	.72701	.74613
3.25	.31953	.49061	.61835	.70649	.76035	.77884
3.50	.34862	.52474	.65168	.73713	.78879	.80668
3.75	.37688	.55647	.68156	.76381	.81305	.83040
4.00	.40422	.58593	.70837	.78706	.83379	.85063
4.25	.43056	.61323	.73240	.80737	.85157	.86792
4.50	.45588	.63849	.75398	.82516	.86684	.88272
4.75	.48016	.66185	.77338	.84077	.88002	.89543
5.00	.50345	.68346	.79083	.85452	.89142	.90637
5.25	.52569	.70342	.80656	.86667	.90134	.91583
5.50	.54689	.72187	.82075	.87744	.91000	.92402
5.75	.56712	.73892	.83358	.88701	.91759	.93114
6.00	.58638	.75469	.84521	.89555	.92427	.93736
7.00	.65444	.80681	.88202	.92183	.94429	.95556
8.00	.70979	.84530	.90764	.93947	.95728	.96686
9.00	.75471	.87409	.92601	.95183	.96619	.97427
10.00	.79119	.89595	.93951	.96079	.97256	.97939
11.00	.82094	.91281	.94969	.96748	.97729	.98308
12.00	.84530	.92601	.95753	.97259	.98088	.98583

The Goldstein function $G(x, \lambda_2)$ $B = 5$

x	0.8	0.85	0.9	0.925	0.950	0.975
$1/\lambda_2$						
0.25	.01876	.01781	.01583	.01427	.01211	.00886
0.50	.07011	.06626	.05866	.05277	.04468	.03266
0.75	.14273	.13422	.11823	.10610	.08962	.06538
1.00	.22503	.21065	.18474	.16544	.13946	.10158
1.25	.30845	.28771	.25151	.22490	.18934	.13776
1.50	.38777	.36085	.31485	.28131	.23666	.17213
1.75	.46036	.42791	.37310	.33330	.25834	.20424
2.00	.52529	.48817	.42578	.38048	.32021	.23306
2.25	.58254	.54172	.47306	.42305	.35634	.25961
2.50	.63262	.58906	.51536	.46140	.38910	.28385
2.75	.67624	.63083	.55324	.49601	.41890	.30605
3.00	.71416	.66768	.58725	.52733	.44610	.32651
3.25	.74714	.70025	.61787	.55582	.47107	.34545
3.50	.77583	.72911	.64555	.58184	.49411	.36308
3.75	.80083	.75475	.67067	.60570	.51547	.37959
4.00	.82266	.77758	.69356	.62769	.53535	.39511
4.25	.84176	.79796	.71448	.64802	.55394	.40979
4.50	.85850	.81622	.73365	.66688	.57139	.42369
4.75	.87321	.83261	.75129	.68445	.58781	.43693
5.00	.88616	.84735	.76756	.70083	.60333	.44955
5.25	.89757	.86062	.78258	.71616	.61802	.46164
5.50	.90766	.87260	.79648	.73054	.63194	.47323
5.75	.91658	.88344	.80938	.74404	.64520	.48437
6.00	.92449	.89325	.82135	.75674	.65781	.49509
7.00	.94829	.92420	.86155	.80075	.70286	.53446
8.00	.96344	.94540	.89206	.83600	.74084	.56919
9.00	.97330	.96011	.91547	.86457	.77326	.60022
10.00	.97986	.97042	.93355	.88787	.80118	.62826
11.00	.98435	.97771	.94756	.90699	.82537	.65378
12.00	.98749	.98291	.95847	.92272	.84643	.67711

Table 1 (continued)

The Goldstein function $G(x, \lambda_2)$ $B = 6$

x	0.2	0.3	0.4	0.5	0.6	0.7
$1/\lambda_2$						
0.25	.00343	.00693	.01093	.01493	.01840	.02072
0.50	.01341	.02685	.04202	.05693	.06960	.07779
0.75	.02915	.05753	.08886	.11899	.14388	.15919
1.00	.04963	.09614	.14606	.19275	.23014	.25186
1.25	.07378	.13991	.20854	.27078	.31904	.34555
1.50	.10067	.18647	.27228	.34758	.40419	.43378
1.75	.12948	.23400	.33448	.41973	.48201	.51322
2.00	.15959	.28120	.39344	.48543	.55095	.58274
2.25	.19043	.32716	.44814	.54407	.61077	.64245
2.50	.22160	.37130	.49831	.59572	.66210	.69315
2.75	.25275	.41325	.54385	.64087	.70574	.73592
3.00	.28363	.45284	.58497	.68016	.74273	.77186
3.25	.31400	.48993	.62196	.71428	.77405	.80205
3.50	.34366	.52459	.65516	.74392	.80059	.82742
3.75	.37253	.55683	.68492	.76969	.82313	.84876
4.00	.40051	.58674	.71159	.79214	.84232	.86679
4.25	.42746	.61443	.73549	.81175	.85876	.88205
4.50	.45338	.64003	.75694	.82892	.87287	.89502
4.75	.47827	.66367	.77619	.84401	.88505	.90609
5.00	.50210	.68549	.79349	.85732	.89561	.91559
5.25	.52485	.70562	.80907	.86908	.90482	.92377
5.50	.54653	.72418	.82311	.87953	.91288	.93085
5.75	.56719	.74130	.83579	.88883	.91999	.93700
6.00	.58684	.75711	.84727	.89713	.92626	.94238
7.00	.65602	.80912	.88353	.92275	.94525	.95821
8.00	.71198	.84727	.90872	.94005	.95776	.96825
9.00	.75711	.87570	.92677	.95220	.96645	.97500
10.00	.79357	.89722	.94005	.96103	.97271	.97978
11.00	.82314	.91379	.95008	.96764	.97738	.98328
12.00	.84727	.92677	.95781	.97272	.98095	.98594

The Goldstein function $G(x, \lambda_2)$ $B = 6$

x	0.80	0.85	0.90	0.925	0.950	0.975
$1/\lambda_2$						
0.25	.02110	.02015	.01802	.01629	.01386	.01018
0.50	.07861	.07479	.06660	.06009	.05102	.03741
0.75	.15929	.15082	.13368	.12032	.10193	.07459
1.00	.24965	.23537	.20778	.18665	.15783	.11533
1.25	.33985	.31941	.28122	.25232	.21313	.15561
1.50	.42418	.39795	.34993	.31383	.26500	.19346
1.75	.50001	.46881	.41226	.36983	.31238	.22817
2.00	.56659	.53149	.46791	.42008	.35515	.25968
2.25	.62427	.58634	.51726	.46496	.39363	.28824
2.50	.67386	.63411	.56095	.50504	.42829	.31418
2.75	.71632	.67567	.59967	.54092	.45964	.33788
3.00	.75263	.71185	.63412	.57320	.48814	.35965
3.25	.78372	.74341	.66486	.60235	.51419	.37979
3.50	.81033	.77101	.69242	.62882	.53814	.39850
3.75	.83319	.79521	.71722	.65295	.56027	.41600
4.00	.85284	.81649	.73961	.67505	.58080	.43245
4.25	.86980	.83525	.75990	.69537	.59993	.44797
4.50	.88446	.85184	.77835	.71411	.61782	.46267
4.75	.89715	.86652	.79516	.73143	.63461	.47664

Table 1 (continued)

The Goldstein function $G(x, \lambda_2)$ $B = 6$

x	0.80	0.85	0.90	0.925	0.950	0.975
5.00	.90818	.87956	.81052	.74751	.65039	.48995
5.25	.91778	.89116	.82457	.76244	.66529	.50268
5.50	.92616	.90148	.83745	.77635	.67936	.51487
5.75	.93348	.91071	.84929	.78931	.69269	.52656
6.00	.93991	.91894	.86017	.80143	.70532	.53780
7.00	.95879	.94414	.89583	.84266	.74996	.57888
8.00	.97043	.96059	.92179	.87469	.78689	.61482
9.00	.97789	.97149	.94089	.89983	.81780	.64673
10.00	.98284	.97884	.95504	.91968	.84390	.67531
11.00	.98626	.98389	.96557	.93544	.86603	.70109
12.00	.98871	.98740	.97343	.94797	.88487	.72446

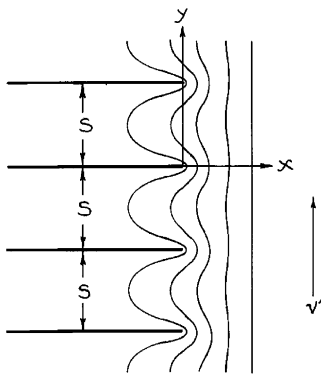


Fig. 7. Flow past a two-dimensional rectilinear grid.

The corresponding circulation if all of the fluid between the lamina were carried along at the same velocity would be $v's$. Consequently, by the two-dimensional analog, the circulation must be reduced by a factor equal to Eq. (3.7.3) divided by $v's$:

$$F = \frac{2}{\pi} \cos^{-1} e^{-f}, \quad (3.7.4)$$

where $f = \pi a/s$.

Applying the factor F to the helicoidal vortex sheets, the distance a from the edge of the sheet is $R-r$ or $R(1-x)$. The spacing s of the sheets at their edges is equal to the linear pitch, $P = 2\pi R\lambda_2$, divided by B , the number of sheets, and multiplied by $\cos \phi$ where $\tan \phi = \lambda_2$. Consequently,

$$f = \frac{B}{2} (1-x) \frac{\sqrt{1+\lambda_2^2}}{\lambda_2}. \quad (3.7.5)$$

Therefore, the circulation function is

$$\frac{B\Gamma\Omega}{2\pi(V+w)w} \cong \frac{Fx^2}{x^2 + \lambda_2^2}. \quad (3.7.6)$$

The Goldstein circulation function is defined by the function displayed as the left-hand side of this equation. Consequently, the right-hand side is an approximation to the Goldstein circulation function:

$$G(x, \lambda_2) \cong \frac{Fx^2}{x^2 + \lambda_2^2}, \quad (3.7.7)$$

where F is given by (3.7.4) and (3.7.5).

Note that by Eqs. (3.7.7) and (3.6.4) Prandtl's F is functionally equivalent to the factor tabulated by Tibery and Wrench and therefore can be taken directly as an approximation to their tabulated function.

The broken lines are the Prandtl approximation to the circulation. The solid lines are the exact solution as presented in Table 1 (Fig. 8).

3.8. The thrust of a propeller with ideal load distribution

The pressure equation for unsteady incompressible potential flow in the absence of an external force such as gravity, which may be neglected in the absence of free surface effects, is

$$\partial\phi/\partial t + u^2/2 + p/\rho = \text{constant}. \quad (3.8.1)$$

Applying this equation to the field of an infinitely long helical vortex system such as that far behind a propeller, both u and $\partial\phi/\partial t$ must approach zero at

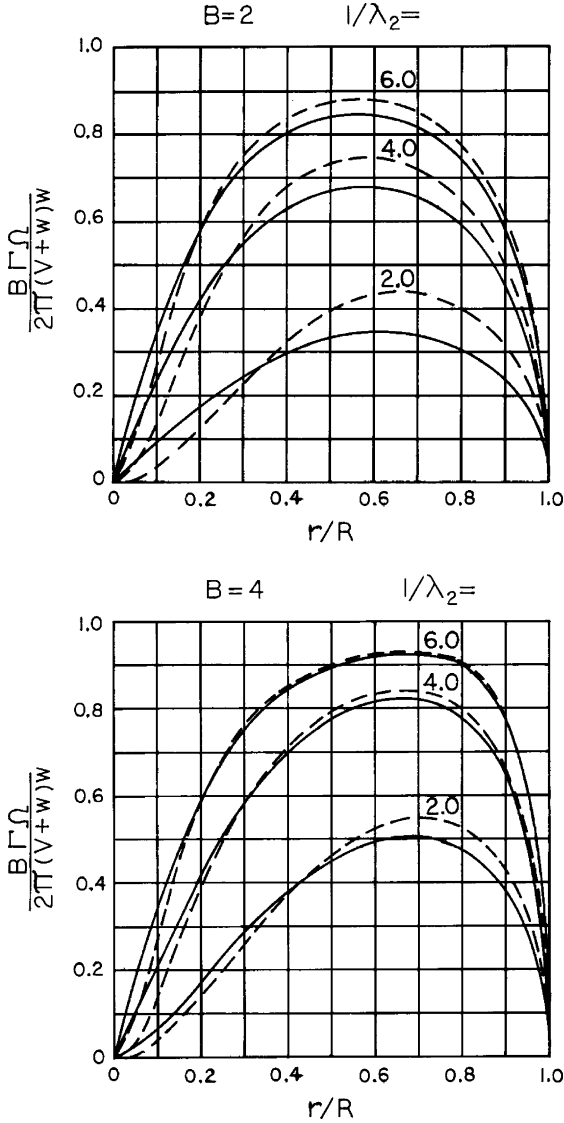


Fig. 8. The ideal radial distribution of circulation.

large distances from the axis. Letting the pressure in the undisturbed fluid at large r be p_∞ , the equation is then

$$\partial\phi/\partial t + u^2/2 + p/\rho = p_\infty/\rho$$

If the entire field pattern moves axially with unchanging form at velocity w in the positive z direction

$$\partial\phi/\partial t = -w\partial\phi/\partial z = -wu_z.$$

Consequently, the pressure equation for such a rigid pattern flow is

$$p - p_\infty + \frac{1}{2}\rho u^2 = \rho w u_z. \quad (3.8.2)$$

By the momentum Eq. (2.2.8), the axial force required to produce the continuous motion of the vortex sheet is

$$T = \int_S \rho(V + u_z)u_z dS + \int_S (p - p_\infty) dS, \quad (3.8.3)$$

where integration is to be performed over a plane normal to the axis of the helicoid and fixed in the undisturbed fluid. Assuming constant density and employing the pressure Eq. (3.8.2),

$$T = \rho \int_S [(V + w)u_z + u_z^2 - u^2/2] dS. \quad (3.8.4)$$

This is a complete expression for the thrust associated with such a vortex system, including the effects of reduced static pressure near the axis.

Theodorsen presented an ingenious evaluation of this expression in terms of κ and ε , two dimensionless quantities that are functions of the pitch of the vortex sheets and can be computed from known values of the Goldstein circulation function. The resulting expression for thrust, Eq. (3.8.14), is derived in the following pages. The values of the functions κ and ε are presented in Tables 2 and 3.

The following considerations enable us to evaluate the first term of Eq. (3.8.4).

On a cylindrical surface, $r_1 = \text{constant}$, consider a triangle ABC drawn so that AB is parallel to the z -axis and spans the distance between two successive helicoidal vortex sheets, Fig. 9. BC is perpendicular to AB. The line CA lies on the surface of one of the sheets. The triangle is imagined to move with the vortex system.

Since no vortex element threads the closed path ABCA,

$$\oint \mathbf{u} \cdot d\mathbf{s} = 0.$$

Since the path of integration moves with the vortex sheets and there is no radial component of vorticity, the velocity along CA is zero. Consequently,

$$\int_A^B u_z dz + \int_B^C -u_\theta r_1 d\theta = 0. \quad (3.8.5)$$

There being screw symmetry, it is evident that u is constant along lines parallel to AC. Therefore, u is distributed along BC exactly as along AB and the integral from A to B may be replaced by an integral from B to C if dz is replaced by

Table 2
The mass coefficient κ

$B =$	2	3	4	5	6
$1/\lambda_2$					
0.25	.00762	.01027	.01239	.01412	.01553
0.50	.02893	.03892	.04678	.05310	.05827
0.75	.06047	.08088	.09670	.1092	.1192
1.00	.09865	.1309	.1552	.1741	.1891
1.25	.1406	.1846	.2170	.2414	.2605
1.50	.1839	.2389	.2781	.3070	.3290
1.75	.2274	.2917	.3362	.3676	.3922
2.00	.2701	.3419	.3903	.4242	.4454
2.25	.3111	.3888	.4396	.4745	.4997
2.50	.3500	.4321	.4842	.5193	.5443
2.75	.3868	.4718	.5243	.5587	.5835
3.00	.4213	.5079	.5601	.5942	.6180
3.25	.4535	.5407	.5922	.6254	.6483
3.50	.4833	.5705	.6209	.6529	.6750
3.75	.5111	.5974	.6465	.6775	.6987
4.00	.5367	.6217	.6695	.6993	.7196
4.25	.5605	.6440	.6901	.7188	.7383
4.50	.5824	.6642	.7087	.7363	.7550
4.75	.6027	.6824	.7255	.7520	.7699
5.00	.6214	.6992	.7407	.7662	.7834
5.25	.6387	.7144	.7545	.7790	.7956
5.50	.6549	.7284	.7671	.7907	.8066
5.75	.6697	.7411	.7785	.8013	.8166
6.00	.6835	.7529	.7891	.8111	.8258
7.00	.7298	.7916	.8233	.8424	.8553
8.00	.7652	.8204	.8484	.8653	.8767
9.00	.7922	.8424	.8676	.8827	.8927
10.00	.8145	.8596	.8826	.8960	.9051
11.00	.8323	.8735	.8942	.9066	.9148
12.00	.8469	.8847	.9037	.9151	.9226

Table 3
Values of ε/κ

$B =$	2	3	4	5	6
$1/\lambda_2$					
0.75	.122	.132	.143	.152	.159
1.00	.179	.198	.215	.231	.242
1.25	.237	.263	.286	.305	.323
1.50	.289	.324	.354	.382	.394
1.75	.334	.380	.414	.441	.485
2.00	.379	.431	.470	.495	.518
2.25	.422	.478	.520	.550	.550
2.50	.459	.520	.563	.596	.619
2.75	.493	.558	.603	.632	.654
3.00	.525	.594	.637	.663	.686
3.25	.556	.624	.667	.696	.715
3.50	.583	.653	.695	.722	.739
3.75	.608	.680	.719	.744	.761
4.00	.632	.701	.740	.765	.781
4.25	.654	.720	.759	.782	.797
4.50	.674	.741	.776	.798	.812
4.75	.693	.757	.791	.812	.825
5.00	.711	.771	.805	.824	.836
5.50	.740	.799	.828	.845	.857
6.00	.766	.821	.847	.863	.873
7.00	.807	.854	.877	.890	.898
8.00	.840	.878	.897	.909	.916
9.00	.862	.897	.912	.923	.930
10.00	.878	.911	.926	.934	.940

$r_1 d\theta \tan \phi$. Multiplying by the number of interleaved vortex sheets, the integrals from B to C become integrals from 0 to 2π and Eq. (3.8.5) becomes

$$\int_0^{2\pi} u_z r_1 d\theta = (1/\tan \phi) \int_0^{2\pi} u_\theta r_1 d\theta. \quad (3.8.6)$$

The relation of bound to trailing vorticity will be discussed in Section 4.1. It will be shown that u_θ is related to bound circulation by

$$B\Gamma(r_0) = \int_0^{2\pi} u_\theta(r_1) r_1 d\theta. \quad (4.1.1)$$

Applying this relation and letting $\tan \phi = \lambda_2/x_1$ we have

$$\int_0^{2\pi} u_z r_1 d\theta = B\Gamma/(\lambda_2/x_1). \quad (3.8.7)$$

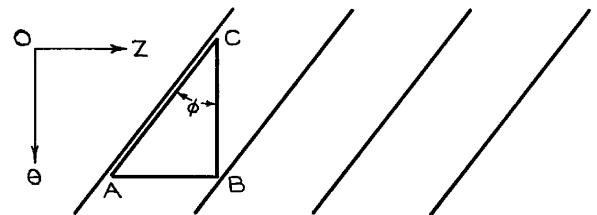


Fig. 9.

The integral needed to evaluate the first term of the thrust Eq. (3.8.4) is

$$\int_S u_z dS = \int_0^{2\pi} \int_0^\infty u_z r_1 dr_1 d\theta.$$

Interchanging the order of integration, applying Eqs. (3.8.7) and (3.6.2), the definition of $G(r_1)$,

$$\begin{aligned} \int_S u_z dS &= \int_0^\infty (B\Gamma/\lambda_2) R x dx \\ &= \pi R_1^2 w \int_0^1 2G(x_1) x_1 dx_1. \end{aligned} \quad (3.8.8)$$

Since $G(r_1) = 0$ when $x_1 \geq 1$, the upper limit of integration has been replaced by unity.

Defining a dimensionless coefficient

$$\kappa = \int_S (u_z/wS) dS. \quad (3.8.9)$$

The first term of the thrust Eq. (3.8.4) is

$$\kappa \rho \pi R_1^2 (V + w)w.$$

By Eq. (3.8.8)

$$\kappa = \int_0^1 2G(x_1)x_1 dx_1. \quad (3.8.10)$$

This relation provides the means to compute the value of κ .

The second term of the thrust Eq. (3.8.4) may be expressed as $\varepsilon \rho \pi R_1^2 w^2$ where ε is a dimensionless coefficient

$$\varepsilon = \int_S u_z^2 dS / \pi R_1^2 w^2 \quad (3.8.11)$$

the evaluation of which will be discussed in Section 3.10.

Now consider the integration of the last term of the thrust equation

$$\int_S u^2 dS.$$

Far behind the propeller the distribution of all variables such as velocity is the same in all planes $z = \text{constant}$ except for a displacement around the axis. The rotational displacement of variables has no effect on a volume integration. Consequently, we can convert the integral over a plane normal to the axis (a Trefftz plane) to a volume integral simply by multiplying by the distance h between two planes z_1 and z_2 . This conveniently simple transformation makes possible an application of Green's theorem which brings us back to a surface integral, but on the surface of the helicoidal vortex sheet. It can be readily evaluated, which was not the case when the integration was over a plane S for which $z = \text{constant}$.

Letting h be the distance between successive vortex sheets, the integral, which is a surface integral on a plane normal to the axis (a Trefftz plane), becomes a volume integral, the volume τ being that contained between two planes $z = z_1$ and $z = z_2$ separated by a distance h :

$$h \int_S u^2 dS = \int_\tau u^2 d\tau = \int_\tau (\nabla \varphi)^2 d\tau.$$

Consider the following form of Green's theorem which transforms a volume integral into an integral over the bounding surfaces [16, pp. 315–316]:

$$\int_\tau [\varphi \nabla^2 \varphi + (\nabla \varphi)^2] d\tau = \int_\sigma \varphi (d\varphi/dn) d\sigma, \quad (3.8.12)$$

where n is the outward pointing normal direction from the surface σ bounding τ .

The bounding surface σ is the complete inner and outer boundary of the region τ . In the case of a body, in this case a vortex sheet moving in an otherwise stationary fluid, the integral over the remote outer boundary $r \rightarrow \infty$ where the velocity is zero vanishes. The integrations over the two adjacent boundaries $z = z_1$ and $z = z_2$ cancel since the field is identical for all z , i.e., the helicoid is assumed to be effectively infinitely long. Consequently, we need only integrate over the inner boundary of σ which is both sides of the helicoidal vortex sheet. Applying Green's theorem we have, since $\nabla^2 \varphi = 0$,

$$h \int_S u^2 dS = \int_\sigma \varphi (d\varphi/dn) d\sigma.$$

Since $d\varphi/dn$ is of equal magnitude and opposite sign on the two sides of the vortex sheet, we may write the last expression as

$$\int_\sigma \Delta \varphi (d\varphi/dn) d\sigma,$$

where $\Delta \varphi$ is the difference in potential between the two sides of the sheet and the integration is now to be performed over just one side of the sheet.

The velocity of the vortex sheet normal to itself is the component of the axial displacement velocity w in the direction of the normal:

$$d\varphi/dn = w \cos \phi$$

and

$$(d\varphi/dn) d\sigma = w \cos \phi d\sigma = w dS,$$

where dS is an element of a plane $z = \text{constant}$. Now

$$\int_S u^2 dS = (1/h) \int_S \Delta \varphi w dS = (1/h) \int_S \Gamma(r) w dS.$$

By a somewhat circuitous route through transformation to a volume integral and a different surface integral, the integral is transformed to a different integral on the original Trefftz plane. Expressing Γ in terms of the Goldstein function

Eqs. (3.6.1) and (3.6.2),

$$\begin{aligned}\int_S u^2 dS &= \int_0^{2\pi} \int_0^R G(r) w^2 r dr d\theta \\ &= \pi R_1^2 w^2 \int_0^1 2G(x_1) x_1 dx_1 \\ &= \pi R_1^2 w^2 \kappa\end{aligned}\quad (3.8.13)$$

and the last term of the thrust equation is just

$$-\frac{1}{2}\rho \int_S u^2 dS = -\frac{1}{2}\rho \pi R_1^2 w^2 \kappa.$$

Summing the terms of the thrust equation, we have

$$T = \kappa \rho \pi R_1^2 V^2 \bar{w} [1 + \bar{w}(\frac{1}{2} + \varepsilon/\kappa)]. \quad (3.8.14)$$

Expressed as a thrust coefficient,

$$K_{T1} = 2\kappa \bar{w} [1 + \bar{w}(\frac{1}{2} + \varepsilon/\kappa)]. \quad (3.8.15)$$

The thrust of the propeller is here expressed entirely in terms of the characteristics of the trailing helicoidal vortex sheets. The relation of the propeller to the helicoidal sheets will be considered in Section 4.

3.9. Efficiency of the propeller with ideal load distribution

The energy passed into the surrounding medium by the propeller is evaluated by consideration of a control volume bounded by the surface S consisting of a plane S_0 normal to the axis far ahead, another S_2 far behind the propeller, and S_1 an everywhere streamwise surface connecting S_0 and S_2 (Fig. 10).

The energy flux outward through S is

$$E = \int_S u_N (\frac{1}{2}\rho u_R^2 + p) dS,$$

where u_N is the velocity in the direction of the outward normal to the surface and u_R is the

resultant velocity on S . Since u_N is zero on S_1 ,

$$\begin{aligned}E &= \int_{S_0} -V(\frac{1}{2}\rho V^2 + p_0) dS_0 \\ &\quad + \int_{S_2} (V + u_z)(\frac{1}{2}\rho u_{R2}^2 + p_2) dS_2,\end{aligned}$$

where

$$u_{R2}^2 = (V + u_z)^2 + u_r^2 + u_\theta^2$$

but since $V dS_0 = (V + u_z) dS_2$

$$\begin{aligned}E &= \int_{S_2} (V + u_z)(\frac{1}{2}\rho u_{R2}^2 - \frac{1}{2}\rho V^2 + p_2 - p_0) dS_2 \\ &= \int_{S_2} (V + u_z)(\frac{1}{2}\rho u^2 + \rho V u_z + p_2 - p_0) dS_2,\end{aligned}$$

where

$$u^2 = u_z^2 + u_r^2 + u_\theta^2.$$

Recalling (3.8.2) this becomes

$$\begin{aligned}E &= \int_{S_2} \rho(V + w)(V + u_z)u_z dS_2 \\ &= \int_S \rho(V + w)(V + u_z)u_z dS,\end{aligned}$$

where S is now the axial projection of the vortex sheets.

Expressing in terms of the integral quantities κ Eq. (3.8.9) and ε Eq. (3.8.11), the energy expenditure of the ideally loaded and frictionless propeller is

$$E = \rho V(V + w)wS(\kappa + \varepsilon w/V). \quad (3.9.1)$$

Expressed as a dimensionless power coefficient

$$K_{P1} = 2\kappa \bar{w}(1 + \bar{w})(1 + \bar{w}\varepsilon/\kappa). \quad (3.9.2)$$

The efficiency is $\eta_i = TV/E = K_{T1}/K_{P1}$. The thrust coefficient K_{T1} is given by Eq. (3.8.15) and we have

$$\eta_i = \frac{1 + \bar{w}(\frac{1}{2} + \varepsilon/\kappa)}{(1 + \bar{w})(1 + \bar{w}\varepsilon/\kappa)}. \quad (3.9.3)$$

3.10. Mass transport in the slipstream

The mass transport through a stationary plane normal to the z -axis is

$$m = \rho \int_S u_z dS.$$

By Eq. (3.8.9) this is just

$$m = \kappa \rho \pi R_1^2 w. \quad (3.10.1)$$

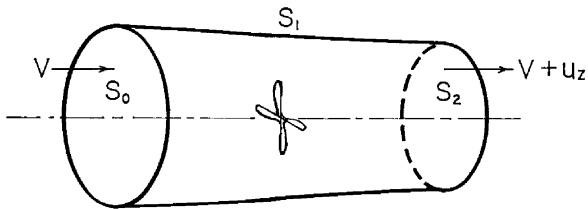


Fig. 10. Control volume for efficiency calculation.

It is clear that the mass flow is equal to that which would be carried backward by a column of fluid equal in diameter to the helical vortex sheets at velocity w , but reduced by the factor $\kappa(B, \lambda_2)$. Consequently, κ is referred to as the mass coefficient.

An upper bound on the value of κ may be established as follows. Recalling Eq. (3.6.2),

$$G(r) = \frac{B\Gamma}{2\pi R_1 w \lambda_2} = \frac{\int_0^{2\pi} u_\theta r d\theta}{2\pi R_1 w \lambda_2}.$$

If we let u_θ have its limiting value as the number of vortex sheets or blades increases indefinitely, we have by Eq. (3.2.7)

$$u_\theta = u_{\theta 1} = w(\lambda_2/x)/(1 + \lambda_2^2/x^2),$$

which is independent of θ . Then

$$G(x) = 1/(1 + \lambda_2^2/x^2). \quad (3.10.2)$$

The limiting value of κ is then, by Eq. (3.8.10),

$$\kappa_\infty = 1 - \lambda_2^2 \ln(1 + 1/\lambda_2^2). \quad (3.10.3)$$

This is not physically very significant in view of the foregoing comments on limits when $B \rightarrow \infty$, but it is an upper bound on κ .

3.11. Evaluation of the axial energy factor ε

The mass transport factor κ is an integral of the axial velocity u_z and the axial energy transport factor ε is an integral of u_z^2 . In view of their common dependence on u_z , it is perhaps not entirely surprising that a simple expression can be developed for $\varepsilon(\lambda_2)$ as a function of $\kappa(\lambda_2)$.

It was shown that the marginal efficiency associated with a small increment of thrust due to an increment of the displacement velocity of the vortex trail $\bar{w}$ is

$$\eta_m = \frac{1}{1 + \bar{w}}. \quad (3.1.6)$$

An expression for the same marginal efficiency can be found from the complete expressions for thrust and power of the propeller with ideal load distribution. The increment of the thrust coefficient in consequence of a small increase $\delta \bar{w}$ in the displacement velocity of the trailing vortex system is

$$\delta K_{T1} = \frac{dK_{T1}}{d\bar{w}} \delta \bar{w} = K'_{T1} \delta \bar{w}$$

and similarly

$$\delta K_{P1} = \frac{dK_{P1}}{d\bar{w}} \delta \bar{w} = K'_{P1} \delta \bar{w},$$

where primes indicate differentiation with respect to $\bar{w}$. The efficiency associated with the small increase of loading on the propeller is then

$$\eta_m = \frac{\delta K_{T1}}{\delta K_{P1}} = \frac{K'_{T1}}{K'_{P1}}. \quad (3.11.1)$$

Differentiating Eqs. (3.8.15) and (3.9.2) and realizing that both ε and κ are functions of $\bar{w}$, we find

$$K'_{T1} = 2\kappa' \bar{w}(1 + \frac{1}{2}\bar{w}) + 2\varepsilon' \bar{w}^2 + 2\kappa(1 + \bar{w}) + 4\varepsilon \bar{w}$$

and

$$K'_{P1} = 2\kappa' \bar{w}(1 + \bar{w}) + 2\varepsilon' \bar{w}^2(1 + \bar{w}) + 2\kappa(1 + 2\bar{w}) + 2\varepsilon(2\bar{w} + 3\bar{w}^2).$$

Equating the two expressions for marginal efficiency, Eqs. (3.1.6) and (3.11.1),

$$\frac{1}{1 + \bar{w}} = \frac{K'_{T1}}{K'_{P1}} \quad (3.11.2)$$

we find

$$\varepsilon = \kappa + \frac{1}{2}\kappa'(1 + \bar{w}) \quad (3.11.3)$$

or

$$\varepsilon = \kappa + \frac{1}{2}\lambda_2 \frac{d\kappa}{d\lambda_2}. \quad (3.11.4)$$

This remarkable relation provides a means for evaluating ε by numerical differentiation of tabulated values of κ . Values of ε are given in Table 3.

4. The propeller related to the vortex trail

4.1. The relation of bound circulation to trailing vorticity

The bound circulation $\Gamma(r_0)$ about an element of the propeller blade is uniquely related to the circulation at a corresponding radius r downstream in the system of helicoidal trailing vortex sheets. This can be seen with the aid of Fig. 11 which depicts a propeller and its trailing vortex sheets.

Consider the line integral of the velocity along the path shown with arrow heads. The path of integration proceeds around a propeller blade and then back on either side of the vortex sheet following the path of vortex elements. At some distance from the propeller, the path of integration

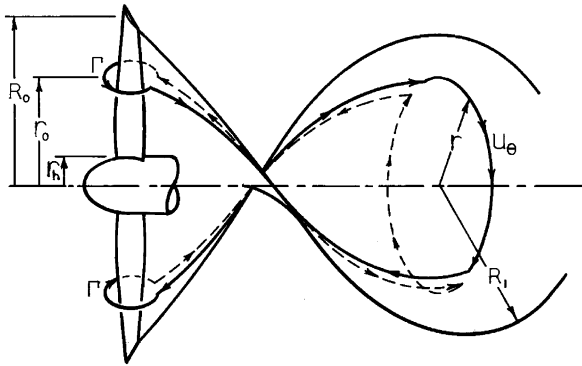


Fig. 11. The propeller and its trailing vortex system.

is completed by circular arcs of radius r . It is evident that the path of integration cuts no vortex sheets and encloses no vortex lines. This statement is verified by the fact that one can imagine the path of integration to be slipped off the vortex system without cutting any vortex lines or tearing any vortex sheets. Therefore, it is found not to enclose any singularities and the line integral of the velocity along the prescribed closed path is necessarily zero.

Around each blade the line integral is the bound circulation Γ . The parts of the path of integration lying along the vortex sheets contribute nothing since the contributions of either side cancel. The integration on the circular path behind the propeller is just the line integral of the tangential component of velocity at radius r_1 . Since the line integral over the whole path is zero it follows that

$$B\Gamma(r_0) = \int_0^{2\pi} u_\theta(r_1)r_1 d\theta. \quad (4.1.1)$$

Consequently,

$$B\Gamma(r_0) = B\Gamma(r_1). \quad (4.1.2)$$

The total bound circulation on the propeller blades at any radius r_0 must be equal to the total shed vorticity within a circle of radius r_1 passing through the vortex filament shed from the elements at r_0 .

It is also evident that the vorticity enclosed within a radius r_1 at any distance z behind the propeller is the same as at any other distance z when r_1 is such as to pass through the same helical vortex filament.

4.2. The effect of a large hub or other central body on circulation distribution

In the foregoing analysis it was found that the circulation on the propeller blades falls continu-

ously on inner parts of the blades, becoming zero only at the axis. This is an acceptable approximation for most aircraft propellers, but there may be cases where there are large spinners and in the case of marine propellers the hub radius is often quite large. In such cases, the circulation is zero at radii less than the hub radius (i.e. inside the spinner) and at larger radii the optimum distribution may be different from the Goldstein distribution.

There have been various attempts to find the optimum distribution of circulation for such propellers by solving the problem of the potential on a system of helicoidal vortex sheets surrounding an infinitely long cylindrical core representing the hub [17,18]. This is a misunderstanding of the problem. Whatever central body there may be, the optimum distribution theorem of Betz still applies to the vortex sheets far behind the propeller where there is no central body. This is the proper understanding of the problem for both pusher and tractor propellers. The concept is further generalized by observing that, neglecting the effects of boundary layers and viscosity in general, the shed vorticity may be displaced behind the propeller by the presence of a nacelle or fuselage, but on passing into the remote wake, must settle into the uniformly translating helicoidal sheet configuration if energy dissipation is to be minimized. For either pusher or tractor propellers, the design problem is, given a system of helicoidal vortex sheets representing the final wake configuration, to find the radial distribution of circulation that must exist before the slipstream contraction and the closure around the hub take place.

The ideal distribution of bound circulation on the propeller blades is such that when the shed vortex system closes behind the hub or a nacelle, its eventual configuration is a set of helicoidal vortex sheets moving uniformly as if rigid, exactly as in the case where there is no central body.

Whether or not there is a hub of significant diameter, the diameter of the propeller is initially not exactly known. Whatever its diameter and hub radius and whatever modification of the flow may exist due to the interference by a nacelle, it is required to find a mapping $\Gamma(x_0)$ on the radial axis of the propeller blade of the circulation distribution $\Gamma(x_1)$ found on the trailing vortex sheets.

Before considering such a solution, some general conclusions may be drawn regarding the circulation distribution when a hub of significant size is present. Since, in the simple case where no hub is considered,

the circulation decreases continuously to zero at the axis, radial displacement of the vortex system requires that the circulation must similarly decrease to zero at the surface of a hub. This is contrary to the result where the ideal fluid model is taken to be a cylindrical inner boundary for the helicoidal vortex sheets. In that case the circulation has a non-zero value at the hub, much like the Goldstein distribution simply truncated at the surface of the hub. This results in a strong vortex around the hub and on the axis in the slipstream. In the case of some torpedo propellers designed in this way [19], the core vortex represented a significant dispersal of energy resulting in a loss of thrust and a loss of efficiency. The low pressure on the hub caused a large vapor cavity extending from the apex of the hub. It is evident that the bound circulation on the blades may be continued to a non-zero value at the hub surface (excepting the boundary layer), but this results in a serious loss of performance and the minimum energy consideration requires the circulation to decrease smoothly to zero at the hub.

A solution to the problem of the ideal distribution of circulation on propeller blades in the presence of a hub is presented in [20]. (see also [21]). A much simplified treatment employing a continuity relation between the flow at the propeller and the ultimate slipstream was found to give nearly identical results and a version of this follows. The assumption of an infinite number of blades justifies the application of an approximate continuity condition. This assumption is only relevant as heuristic justification of the use of a continuity condition and is otherwise not invoked. The Goldstein distribution of circulation on the trailing vortex system remains in effect.

A simple continuity relation may be written between the flow through the propeller and in the slipstream:

$$(V + u_{z_0})2\pi r_0 dr_0 = (V + u_{z_1})2\pi r_1 dr_1.$$

Making the assumption that the ratio of the two parenthetical velocity terms is approximately a constant m , we have

$$m \int_{r_h}^{r_0} r_0 dr_0 = \int_0^{r_1} r_1 dr_1$$

or

$$m(R_0/R_1)^2 \int_{x_h}^{x_0} x_0 dx_0 = \int_0^{x_1} x_1 dx_1.$$

Hence,

$$x_0^2 = x_h^2 + x_1^2(R_1/R_0)^2/m.$$

The boundary condition $x_0 = x_1 = 1$ gives

$$(R_1/R_0)^2/m = 1 - x_h^2$$

and therefore

$$x_0^2 = x_h^2 + x_1^2(1 - x_h^2). \quad (4.2.1)$$

This expression relates the dimensionless radial coordinate x_1 of a trailing vortex element and the coordinate x_0 of its origin at the propeller plane without knowledge of the radius R_0 of the propeller. It applies whether or not there is a hub of significant diameter.

Fig. 12 shows the ideal radial distribution of circulation on three propellers designed for the same conditions and having, respectively, no hub, hub radius $x_h = .2$, and $x_h = .4$. Since the circulation is plotted against r/R , it falls to zero at the same point for all three propellers, but the propellers with hubs will have a slightly greater diameter.

4.3. The velocities at the propeller blade

Having defined the configuration of the trailing vortex sheet and found the associated thrust and torque, it remains to determine the configuration of the propeller which gives rise to such a trailing vortex system. To this end it is necessary to establish the diameter of the propeller and the components of velocity and the circulation at each section of the

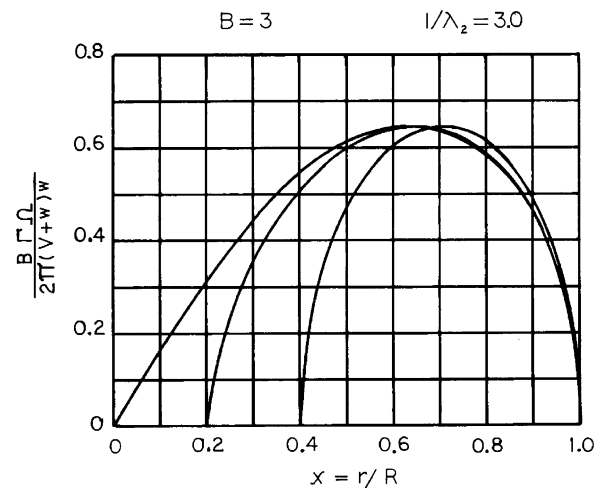


Fig. 12. The distribution of circulation on propellers without and with hubs.

propeller blades. With this information at hand, the geometry of blade sections can be established.

The theory of aircraft propellers, following the original development of finite wing theory, has nearly always proceeded as a lifting line analysis. That is, blade elements are assumed to lie on radial lines and may be considered to act as two-dimensional foils upon which the forces are the same as would be found in a uniform two-dimensional flow with the same local velocity and direction. For this to be justifiable, the velocity field must be effectively uniform in the immediate region of the airfoil. Aircraft propeller blades are almost always narrow enough that this assumption is reasonable. It is possible to develop a correction to the camber of blade elements to compensate for the curvature of the velocity field, but this refinement is probably not worthwhile for typical aircraft propellers. Marine propellers, on the other hand, usually have wide blades whose developed area may be as great as the disc area of the propeller and the blades may incorporate a large sweep. Consequently, lifting line assumptions are not justified in marine applications and there has been a great amount of analytical work on marine propellers employing lifting surface methods. See for example [22].

From the following argument it can readily be seen that the induced velocities at the propeller plane tend to be half the induced velocity at a corresponding point on the helicoidal vortex sheet far behind the propeller:

Assume a set of equally spaced right helicoidal vortex sheets extending in both directions from a plane normal to the axis. Consider any point on the vortex sheets where they intersect the plane. From the Biot–Savart law, it can be seen that the induced velocity at such a point due to a vortex element at an arbitrary distance from the plane is exactly equal and in the same sense as the velocity induced by a like element at the same distance in the other direction from the plane. (See Appendix B). Consequently, if the helicoidal vortices are semi-infinite, extending in only one direction from the plane, the velocities on the plane will be half what they would be for the doubly infinite system. This is taken as an adequate approximation for the velocities induced at the propeller plane by the trailing vortex system except that the tangential velocity u_{θ_0} is modified for the effect of radial displacement of the trailing vortex system immediately behind the propeller.

It must be recognized that representing the vortex system behind the propeller by regular semi-infinite helicoidal vortex sheets is a simplification since both the pitch and the radius of the vortices will be modified to some extent immediately behind the propeller. Also, it was pointed out in Section 3.5 that the helicoidal sheets are unstable and at some distance behind the propeller will roll up into a set of helical vortex filaments, one for each blade, and another of opposite sense on the axis. It was shown that the rolling up of the sheets at a distance from the propeller has no significant effect on the velocity field at the propeller, but the contraction of the trailing vortex system immediately behind the propeller must be taken into account. The exception to this is the case of a lightly loaded propeller where a simplified treatment is appropriate.

The radial displacement of the trailing vortex system immediately behind the propeller occurs in any case and is augmented by the effect of a hub of significant size. The effect of the radial displacement is taken into account by observing that the circulation as measured by a line integral on a circle of radius r must be the same at any plane behind the propeller when r is drawn through the same vortex filament. That is, ru_{θ} is constant.

Consequently, the tangential velocity at the propeller is related to the velocity at the trailing vortex system by

$$u_{\theta_0}r_0 = \frac{1}{2}u_{\theta_1}r_1 \quad (4.3.1)$$

or

$$\bar{u}_{\theta_0}x_0R_0 = \frac{1}{2}\bar{u}_{\theta_1}x_1R_1$$

and

$$\bar{u}_{\theta_0} = \frac{1}{2}\bar{u}_{\theta_1}(x_1/x_0)(R_1/R_0).$$

The relation between the dimensionless coordinates x_1 and x_0 is given by Eq. (4.2.1) and the ratio R_1/R_0 of the helicoidal vortex trail diameter to the propeller diameter will be developed in the next section.

The velocities u_{z_1} and u_{θ_1} are given by Eqs. (3.2.6) and (3.2.7). Now we have

$$\bar{u}_{\theta_0} = \frac{1}{2}\bar{w}(1 + \bar{w})(\lambda/x_0)/(1 + \lambda_2^2/x_1^2) \quad (4.3.2)$$

and

$$\bar{u}_{z_0} = \frac{1}{2}u_{z_1} = \frac{1}{2}\bar{w}/(1 + \lambda_2^2/x_1^2). \quad (4.3.3)$$

The magnitude of U_0 , the relative velocity at a blade element, is required in order to find the blade chord and angle of attack corresponding to the

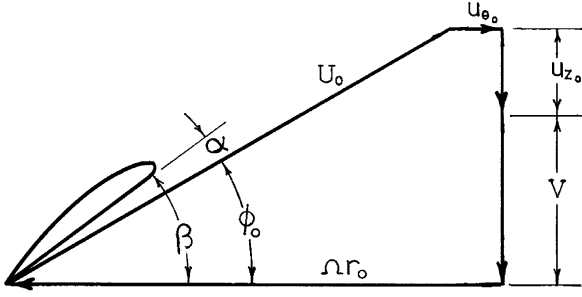


Fig. 13. Velocities at a blade element.

known bound circulation. Referring to Fig. 13,

$$U_0^2 = (V + u_{z_0})^2 + (\Omega r_0 - u_{\theta_0})^2, \quad (4.3.4)$$

$$(U_0/V)^2 = (1 + \bar{u}_{z_0})^2 + (x_0/\lambda - \bar{u}_{\theta_0})^2,$$

where $\bar{u}_{\theta_0}$ and $\bar{u}_{z_0}$ are given by Eqs. (4.3.2) and (4.3.3).

The pitch angle of the relative wind at a blade element is

$$\tan \phi_0 = \frac{V + u_{z_0}}{\Omega r_0 - u_{\theta_0}} = \frac{1 + \bar{u}_{z_0}}{x_0/\lambda - \bar{u}_{\theta_0}}. \quad (4.3.5)$$

4.4. The propeller diameter

Since the slipstream behind the propeller contracts, the first objective is to determine the diameter of the propeller. The ratio of the radius of the propeller to the radius of the assumed trailing vortex system can be deduced by equating the thrust generated by the bound circulation on the propeller, Eq. (2.1.2), to the thrust implied by the backward motion of the vortex system, Eq. (3.8.14). These equations express the thrust in terms of the respective propeller and vortex trail radii. Since these two quite different expressions describe the same dynamic system, their agreement must implicitly define the relative radii of the propeller and the trailing vortex sheets.

The circulation distribution on the trailing vortex system is, Eq. (3.6.2)

$$B\Gamma = 2\pi R_1 w \lambda_2 G(x_1) = 2\pi R_0 V \bar{w}(1 + \bar{w}) \lambda G(x_1). \quad (4.4.1)$$

Substituting this expression in Eq. (2.1.2), the thrust is

$$T_0 = \rho \pi R_0^2 V^2 \bar{w}(1 + \bar{w}) \int_{x_h}^1 2G(x_1)(x_0 - \lambda \bar{u}_{\theta_0}) dx_0. \quad (4.4.2)$$

Applying Eq. (4.2.1),

$$\int_{x_h}^1 2G(x_1)x_0 dx_0 = (1 - x_h^2) \int_0^1 2G(x_1)x_1 dx_1 = (1 - x_h^2)\kappa.$$

With the aid of Eqs. (4.2.1) and (4.3.2) we find

$$\int_{x_h}^1 2G(x_1)\bar{u}_{\theta_0} dx_0 = \frac{1}{2}\bar{w}(1 + \bar{w})\lambda I_1,$$

where

$$I_1 \equiv \int_0^1 \frac{2G(x_1)x_1^3 dx_1}{(x_1^2 + \lambda_2^2)(x_1^2 + c)} \quad (4.4.3)$$

and

$$c = x_h^2/(1 - x_h^2)$$

and finally

$$T_0 = \rho \pi R_0^2 V^2 \bar{w}(1 + \bar{w})[\kappa(1 - x_h^2) - \frac{1}{2}\bar{w}(1 + \bar{w})\lambda^2 I_1]. \quad (4.4.4)$$

Equating T_0 to the thrust associated with the trailing vorticity Eq. (3.8.14), we find the ratio of the propeller radius to the radius of the vortex trail:

$$(R_0/R_1)^2 = \frac{[1 + \bar{w}(\frac{1}{2} + \varepsilon/\kappa)](1 + \bar{w})}{(1 - x_h^2) - \frac{1}{2}\bar{w}(1 + \bar{w})\lambda^2 I_1/\kappa}. \quad (4.4.5)$$

4.5. Lift coefficient and blade angle

Recalling Eqs. (4.1.1) and (4.1.2), the lift coefficient and the chord of a blade element at r_0 must be such that the bound circulation is equal to the total shed vorticity within a circle of radius r_1 passing through the vortex filament shed from the blade element at r_0 .

The lift of a blade element dr is

$$dL = c_l \frac{\rho}{2} U_0^2 c dr = \rho \Gamma U_0 dr,$$

hence the bound circulation is

$$B\Gamma = \frac{1}{2}c_l Bc U_0 = c_l \sigma \pi R U_0 \quad (4.5.1)$$

recalling Eq. (3.6.2), the corresponding circulation in the trailing vortex system is

$$B\Gamma = 2\pi R_1 w \lambda_2 G(x) = 2\pi R \lambda w(1 + \bar{w})G(x).$$

Equating this to (4.5.1), we have

$$\sigma c_l = 2\lambda \bar{w}(1 + \bar{w})G(x)/(U_0/V). \quad (4.5.2)$$

Typically, c_l will be chosen such that c_d/c_l is minimized. The exceptions are those conditions where U_0 is so high that there may be an excessive drag rise due to compressibility in which case the lift coefficient must be reduced. Having chosen the lift coefficient, the angle of attack α , the solidity σ and the chord are established. The blade angle β is then just $\beta(x) = \alpha + \phi_0$.

4.6. Thrust and torque costs of profile drag

The optimum distribution of bound circulation was determined without consideration for the existence of profile drag. Profile drag may be treated as an additive force without modification of the distribution of trailing vorticity because it acts in a direction normal to the lift force and, to the first order, does not affect the form of the trailing vortex system.

The loss of thrust in consequence of the profile drag of the blade sections relative to the thrust of the ideally loaded frictionless propeller is

$$\begin{aligned} dT_p &= -c_d \frac{\rho}{2} U_0^2 B c dr \sin \phi_0 \\ &= -c_d \rho U_0^2 \sigma \pi R^2 \sin \phi_0 dx, \end{aligned}$$

$$dK_T = -2c_d \sigma (U_0/V)^2 \sin \phi_0 dx$$

and the contribution to the thrust coefficient is

$$\Delta K_T = -2 \int_0^1 c_d \sigma (U_0/V)^2 \sin \phi_0 dx. \quad (4.6.1)$$

The torque due to profile drag is

$$\begin{aligned} dQ_p &= c_d \frac{\rho}{2} U_0^2 B c r dr \cos \phi_0 \\ &= c_d \rho U_0^2 \sigma \pi R^3 \cos \phi_0 x dx, \end{aligned}$$

$$dK_Q = c_d 2\sigma (U_0/V)^2 x dx \cos \phi_0$$

and the contribution to the torque coefficient is

$$\Delta K_Q = 2 \int_0^1 c_d \sigma (U_0/V)^2 \cos \phi_0 x dx \quad (4.6.2)$$

the corresponding effect on the power coefficient is $\Delta K_P = \Delta K_Q/\lambda$.

The angle ϕ_0 is given by

$$\tan \phi_0 = \frac{V + u_{z_0}}{\Omega r_0 - u_{\theta_0}} = \frac{1 + \bar{u}_{z_0}}{x_0/\lambda - \bar{u}_{\theta_0}}. \quad (4.3.5)$$

Consequently, the thrust and power coefficients for the propeller with ideal load distribution with allowances for the effects of profile drag of the blade sections on thrust and torque are, from Eqs. (3.4.12)

and (4.6.1)

$$K_T = K_{T1}/(R/R_1)^2 + \Delta K_T \quad (4.6.3)$$

from Eqs. (3.5.2) and (4.6.2)

$$K_P = K_{P1}/(R/R_1)^2 + \Delta K_P. \quad (4.6.4)$$

The efficiency of the propeller is

$$\eta = TV/P = K_T/K_P. \quad (4.6.5)$$

5. Design and performance computations

5.1. Design procedure for a propeller with ideal load distribution

Since the theory of the propeller with ideal load distribution has been developed from conditions on the trailing vortex system, the design of a propeller proceeds from specification of the configuration of the trailing vortex system, which requires the specification of B , the number of blades, and only two parameters, the advance ratio λ_1 and $\bar{w}$ the relative displacement velocity of the helicoidal sheets. In the case of a practical design problem, it is somewhat awkward to have to start with these parameters which are remote from immediate engineering requirements, but approximate values of λ_1 and $\bar{w}$ can be estimated from more common engineering requirements.

Starting with required values of λ and the thrust coefficient K_T , $\bar{w}$ is obtained by solving Eq. (3.8.15) for $\bar{w}$:

$$\bar{w} = \frac{-1 + \sqrt{1 + K_{T1}(1 + 2\varepsilon/\kappa)/\kappa}}{1 + 2\varepsilon/\kappa}. \quad (5.1.1)$$

For the first estimate of $\bar{w}$ let $K_{T1} = K_T$ and let $\lambda_2 = \lambda$. Then κ is obtained from Table 2 and ε/κ from Table 3. The resulting value of $\bar{w}$ from Eq. (5.1.1) makes available a new value of λ_2 since $\lambda_2 = \lambda_1(1 + \bar{w})$. Then improved values of the functions κ and ε/κ are available from the tables and $\bar{w}$ may be refined. The input may be adjusted and the design process repeated as necessary until the advance ratio λ and thrust coefficient K_T are in accord with the design requirements.

The thrust and power coefficients and the efficiency associated with the optimum circulation distribution and without viscous losses are then

$$K_{T1} = 2\kappa\bar{w}[1 + \bar{w}(\frac{1}{2} + \varepsilon/\kappa)], \quad (3.8.15)$$

$$K_{P1} = 2\kappa\bar{w}(1 + \bar{w})(1 + \bar{w}\varepsilon/\kappa), \quad (3.9.2)$$

$$\eta_i = \frac{1 + \bar{w}(\frac{1}{2} + \varepsilon/\kappa)}{(1 + \bar{w})(1 + \bar{w}\varepsilon/\kappa)}. \quad (3.9.3)$$

The radius of the propeller is then related to the radius of the helicoidal vortex sheets by the following expression:

$$(R_0/R_1)^2 = \frac{[1 + \bar{w}(\frac{1}{2} + \varepsilon/\kappa)]/(1 + \bar{w})}{(1 - x_h^2) - \frac{1}{2}\bar{w}(1 + \bar{w})\lambda^2 I_1/\kappa}, \quad (4.4.5)$$

where

$$I_1 \equiv \int_0^1 \frac{2G(x_1)x_1^3 dx_1}{(x_1^2 + \lambda^2)(x_1^2 + c)} \quad (4.4.3)$$

and

$$c = x_h^2/(1 - x_h^2).$$

In Eq. (4.4.3) the Goldstein function $G(x)$ must be taken from the Table 1. Consequently, I_1 must be obtained by numerical integration.

When the hub is small enough to have negligible effect we set $x_h = 0$ and Eq. (4.4.5) reduces to

$$(R_0/R_1)^2 = \frac{[1 + \bar{w}(\frac{1}{2} + \varepsilon/\kappa)]/(1 + \bar{w})}{1 - \frac{1}{2}\bar{w}(1 + \bar{w})\lambda^2 I_1/\kappa}, \quad (5.1.2)$$

where

$$I_1 = \int_0^1 \frac{2G(x_1)x_1 dx_1}{x_1^2 + \lambda^2}.$$

We now have the advance ratio of the propeller, $\lambda = \lambda_1/(R_0/R_1)$.

The relative velocity at a blade element $U_0(x)$ is given by

$$(U_0/V)^2 = (1 + \bar{u}_{z_0})^2 + (x_0/\lambda - \bar{u}_{\theta_0})^2, \quad (4.3.4)$$

where $\bar{u}_{\theta_0}$ and $\bar{u}_{z_0}$ are given by Eqs. (4.3.2) and (4.3.3).

The pitch angle of the relative wind at a blade element is

$$\tan \phi_0 = \frac{V + u_{z_0}}{\Omega r_0 - u_{\theta_0}} = \frac{1 + \bar{u}_{z_0}}{x_0/\lambda - \bar{u}_{\theta_0}}. \quad (4.3.5)$$

The radial distribution of blade chord and lift coefficient are given by

$$\sigma c_l = 2\lambda\bar{w}(1 + \bar{w})G(x)/(U_0/V). \quad (4.5.2)$$

The lift coefficient c_l will usually be selected so as to minimize the ratio c_d/c_l , but a lesser value may be advisable where the local Mach number is so high as to raise the possibility of compressibility drag rise or excessive noise. Having selected the lift coefficient,

the angle of attack is $\alpha = c_l/a_0 + \alpha_{L_0}$ and the blade angle is $\beta = \phi_0 + \alpha$.

Having selected the lift coefficient, the corresponding σ determines the local blade width.

The contributions of the section profile drag to the thrust and torque coefficients are

$$\Delta K_T = -2 \int_0^1 c_d \sigma (U_0/V)^2 \sin \phi_0 dx, \quad (4.6.1)$$

$$\Delta K_Q = 2 \int_0^1 c_d \sigma (U_0/V)^2 \cos \phi_0 x dx, \quad (4.6.2)$$

$$\Delta K_P = \Delta K_Q/\lambda. \quad (5.1.3)$$

Finally, the thrust and power coefficients are

$$K_T = K_{T1}/(R/R_1)^2 + \Delta K_T, \quad (4.6.3)$$

$$K_P = K_{P1}/(R/R_1)^2 + \Delta K_P. \quad (4.6.4)$$

The efficiency of the propeller is

$$\eta = K_T/K_P. \quad (4.6.5)$$

5.2. Performance of a given propeller

The foregoing treatment of the propeller with ideal radial distribution of load is, of course, not directly applicable to the computation of the performance of a given propeller nor to a propeller designed for ideal load distribution when operating at other than design conditions. A solution for the performance at non-ideal conditions can be constructed if we accept a degree of approximation for the velocity field at the blade elements.

The classical solution known as the combined blade element and momentum theory [7] was based on an assumed independence of blade elements. This conception can only be justified for an infinite bladed propeller where the flow may be considered to be confined to concentric annular stream tubes. The thrust and torque contribution of each blade element dr was computed by combining two-dimensional section characteristics and the local velocity at the blade element computed by consideration of the angular and axial momentum imparted to the corresponding annular stream tube.

In the light of an understanding of the trailing vortex system, there can be no independence of elements as assumed in the combined blade element-momentum theory. The trailing vorticity

shed from each blade element will contribute to the perturbation velocity at every other blade element. The error inherent in the momentum theory is particularly egregious near the tips of the propeller blades. However, the combined theory was useful when corrections by means of the Prandtl factor (see Section 3.7) were made near the tips.

A computational solution to the loading of a given propeller employing vortex concepts is presented here. In the case of a propeller of given geometry the displacement velocity of the helical shed vortices from each element is generally not the same for all elements as in the ideally loaded propeller. The velocity induced at each blade element will be assumed to be that which would occur if there were a helicoidal trailing vortex system with uniform displacement velocity equal to the displacement velocity consistent with the circulation at that element. This is, to a degree, a reversion to the “independence of blade elements”, but is not so thoroughgoing as in the momentum theory. It can be argued that it is justifiable because the velocities induced by the shed vorticity of adjacent elements will be similar for any propeller with geometry varying smoothly with radius. Perhaps more convincingly, it is justified by satisfactory experience with computations of this sort [23].

Given the advance ratio λ and the geometric description of the propeller: the number of blades B , the blade angle $\beta(x)$, chord $c(x)$, and section characteristics a_0 , α_{L_0} , c_d , it is required to find the thrust and torque of the propeller.

The general procedure is to find $\bar{w}(x)$ at a number of stations such that the induced velocity and consequent lift coefficient implied by local two-dimensional flow conditions at an element are consistent with the circulation that follows from the value of $\bar{w}(x)$. The proper value will be found by an iterative procedure.

Eq. (4.5.2) is the circulation condition and establishes the local section lift coefficient as a function of $\bar{w}$ and the corresponding Goldstein function $G(x)$. The Goldstein function is also dependent on $\bar{w}$ since it is a function of $\lambda_2 = \lambda_1(1 + \bar{w})$:

$$\sigma c_l = 2\lambda\bar{w}(1 + \bar{w})G(x)/(U_0/V). \quad (4.5.2)$$

We now let $\bar{w} = \bar{w}(x)$ and apply Eq. (4.5.2) at a number of blade elements. This is inconsistent with the original definition of $\bar{w}$ as the displacement

velocity of an undeforming helicoidal vortex sheet, but facilitates the approximate evaluation of the induced velocities at each blade element. Recalling Eq. (4.3.4)

$$(U_0/V)^2 = (1 + \bar{u}_{z_0})^2 + (x_0/\lambda - \bar{u}_{\theta_0})^2, \quad (4.3.4)$$

where $\bar{u}_{\theta_0}$ and $\bar{u}_{z_0}$ are given by Eqs. (4.3.2) and (4.3.3).

The pitch angle of the relative wind at a blade element is

$$\tan \phi_0 = \frac{V + u_{z_0}}{\Omega r_0 - u_{\theta_0}} = \frac{1 + \bar{u}_{z_0}}{x_0/\lambda - \bar{u}_{\theta_0}}. \quad (4.3.5)$$

The expressions for $\bar{u}_{\theta_0}$ and $\bar{u}_{z_0}$ are functions of $\lambda_2 = \lambda_1(1 + \bar{w})$ whereas Eq. (4.5.2) contains λ . For present purposes it is sufficient to assume that $\lambda_1 = \lambda$, that is, to neglect the usually slight contraction of the vortex trail immediately behind the propeller.

Usually $x_0 = x_1$, but in the case where there is a large hub or spinner, the circulation distribution at the propeller is displaced outward compared with the distribution in the trailing vortex as indicated by Eq. (4.2.1):

$$x_0^2 = x_h^2 + x_1^2(1 - x_h^2) \quad (4.2.1)$$

and

$$(R_0/R_1)^2 = 1/(1 - x_h^2).$$

The local two-dimensional flow condition requires that the lift coefficient be

$$c_l = a_0(\beta - \phi_0 - \alpha_{L_0}), \quad (5.2.1)$$

where ϕ_0 is given by Eq. (4.3.5).

At each radial station a value of $\bar{w}$ must be found such that Eqs. (4.5.2) and (5.2.1) are satisfied simultaneously, i.e. yield the same value of c_l . As a practical computation they may be satisfied by an iterative procedure at each of ten or more radial stations. The thrust and torque coefficients are then

$$K_T = 2 \int_0^1 (c_l \cos \phi_0 - c_d \sin \phi_0) \sigma (U_0/V)^2 dx, \quad (5.2.2)$$

$$K_Q = 2 \int_0^1 (c_l \sin \phi_0 + c_d \cos \phi_0) \sigma (U_0/V)^2 x dx \quad (5.2.3)$$

and the efficiency is

$$\eta = \lambda K_T / K_Q. \quad (5.2.4)$$

6. Propeller interaction with a body

The interaction or interference between a propeller and another aerodynamic body is usually very complicated, but some useful generalizations can be developed by consideration of an actuator disc as a representation of a propeller. Two general types of interference may be recognized: (a) the propeller subject to a locally increased or reduced velocity due to an adjacent body where the flow is essentially a potential flow, i.e. the total pressure is constant, and (b) running in the wake of a body where the total pressure is reduced by viscous effects. The two effects are very different. In practice, the two types of interference may both exist and, of course, the velocity perturbations will not be uniform over the propeller disc, but the general nature of the interference can be elucidated by considering an impulse disc running in a uniformly perturbed stream. The fundamental aerodynamics of the impulse disc is outlined in Appendix A.

6.1. Interaction with a large body

We first consider the case of an impulse disc operating in a velocity field locally modified by the presence of a large body. No viscous wake flows into the propeller and for present considerations the body is assumed not to be subject to viscous effects. The nomenclature for the flow through the propeller is indicated in Fig. 14. S_0 is the area of the impulse disc and S is the area of a section of the slipstream at a distance from the propeller.

The velocity through the disc in the absence of interference would be $(V + u_{z0})$. It is assumed that the only effect of the interfering body is to modify the velocity at the disc by a factor m . The disturbance of the flow through the propeller is local and leaves the final slipstream velocity $(V + u_z)$ unaffected. Consequently, the total force on the system of propeller and interfering body is given by the momentum Eq. (2.2.8):

$$T_{\text{net}} = \rho(V + u_z)u_z S. \quad (6.1.1)$$

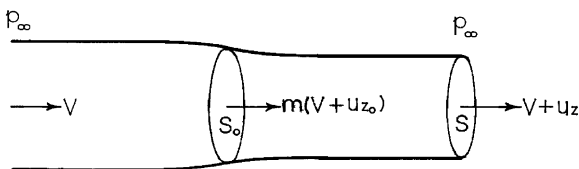


Fig. 14. Flow through a propeller in a region of increased velocity.

The increase of total pressure applied as an increase in static pressure across the impulse disc is

$$\Delta p = \frac{\rho}{2}[(V + u_z)^2 - V^2] = \rho u_z(V + u_z/2). \quad (6.1.2)$$

The area of the disc is related to the slipstream cross-section area by the continuity equation

$$S(V + u_z) = S_0 m(V + u_{z0}). \quad (6.1.3)$$

It is shown in Appendix A that the velocity induced at an impulse disc is half of the final velocity imparted, that is $u_{z0} = u_z/2$.

The actual thrust of the disc is now

$$T = \Delta p S_0 = \rho u_z(V + u_z)S/m. \quad (6.1.4)$$

Consequently,

$$T_{\text{net}}/T = m. \quad (6.1.5)$$

This expresses the ratio of the total thrust on the system (body + propeller) to the actual thrust of the propeller. Consequently, if the propeller is running in a region of increased velocity we have $m > 1$ and $T_{\text{net}} > T$ and there must be a forward thrust on the body. Conversely, if the propeller is running in a region of reduced velocity there will be a drag force on the body and the total thrust of the system will be less than the thrust of the propeller.

The propulsive efficiency of the propeller in the presence of an interfering body is

$$\eta = \frac{T_{\text{net}} V}{T m(V + u_{z0})} = \frac{V}{V + u_{z0}}. \quad (6.1.6)$$

Expressed in terms of a thrust coefficient where the thrust is the net thrust of the propeller and interfering body

$$\eta = \frac{2}{1 + \sqrt{1 + K_T/m}}. \quad (6.1.7)$$

It is evident that placing a propeller in a region of increased velocity increases the net propulsive efficiency, a consequence of working on a larger mass flow of air. The converse is also true if the propeller is working in a region of reduced velocity.

6.2. Interaction of a tractor propeller with a nacelle or fuselage

The impulse disc treatment of a propeller in a perturbed velocity field in the preceding section made possible some very general conclusions on interference effects, but gives no specific design information. In the following we consider the more

specific problem of the design of a tractor propeller at the nose of a nacelle or fuselage.

The effect of a nacelle or fuselage on the distribution of the loading on a propeller for maximum efficiency can be developed from the requirement that the trailing vortex system be a helicoidal sheet moving as if rigid, exactly as in the case of an isolated propeller. First consider an ideally loaded propeller moving in free air without interference from any adjacent body. At some distance behind the propeller the trailing vortices appear as a regular helicoidal sheet. Now, at some lesser distance behind the propeller, interpose on its axis a streamlined nacelle. The nacelle, being at a sufficient distance, has no effect on the propeller. Neglecting viscous effects and the instability of vortex sheets, it also has no effect on the final form of the vortex system, which will flow around the nacelle and finally resume its fixed helicoidal form.

Now consider how the propeller must be modified if it is moved downstream to a position immediately in front of the nacelle and is required to give rise to the same final form of the trailing vortex system, the remote helicoidal trailing vortex sheet being regarded as an unchanging given (Fig. 15). The flow in front of the nacelle will be retarded and there will be a radial displacement of streamlines. As it is moved to proximity to the nacelle, blade elements of the free running propeller must be displaced radially and the bound circulation of each element must remain unchanged if the final trailing vortex system is to remain unchanged.

Since, in locating the propeller close to the nacelle, the relative peripheral velocity at a blade element $\Omega r - u_\theta$ is subject to little change while the axial component $V + u_z$ may be substantially reduced by an additional interference from the nacelle, the angle of attack and the circulation will be increased unless the local blade angle β is reduced. The design of a propeller in the presence of a nacelle with ideal load distribution requires the determination of the radial coordinates of blade

elements in relation to the radii of the hypothesized free-running propeller and the determination of the blade angle β which results in the proper bound circulation.

The flow around the nacelle may be described by a distribution of sources and sinks on the axis. However, the flow in the region of a propeller just ahead of a nacelle or fuselage is probably adequately represented by a single source.

The transformation of the design of a free-running propeller to a propeller at the nose of a nacelle will result in the stretching of the circulation distribution over a greater radius. This will usually result in a somewhat greater thrust, but both propellers result in the same trailing helicoidal vortex system, hence the same net thrust. The difference is due to a drag force on the nacelle induced by the proximity of the propeller. We may also observe that the design of a pusher propeller with ideal load distribution is, if we neglect the effects of viscosity, exactly the same as for a tractor propeller.

The radial displacement of streamlines near the nacelle can be estimated by invoking a continuity condition as in the case of the large hub or spinner in Section 5.2. This amounts to employing a Stokes' stream function and is properly applied only in the case of a propeller with $B \rightarrow \infty$ in which case the flow is confined to annular stream tubes. However, it is an adequate tool for the present problem.

Letting unsubscripted variables refer to the hypothetical upstream free-running propeller, we write

$$\Psi = \int_0^r (V + u_{z0})r \, dr = \int_0^{r_0} (V + u_{z0} + u_{zN})r_0 \, dr_0, \quad (6.2.1)$$

where u_{zN} is the axial component in the plane of the propeller due to the presence of the nacelle. If we make the approximation that $(V + u_{z0})$ may be considered to be constant, that is independent of r , this becomes

$$(V + u_{z0})(r^2 - r_0^2)/2 = \int_0^{r_0} u_{zN}r_0 \, dr_0. \quad (6.2.2)$$

This equation relates the radial coordinate r_0 of each element on the propeller near the nacelle to its corresponding element at radius r on the hypothetical free-running propeller. The integral may be carried out in any way suitable to the particular problem.



Fig. 15. A free running propeller and the equivalent propeller on a nacelle.

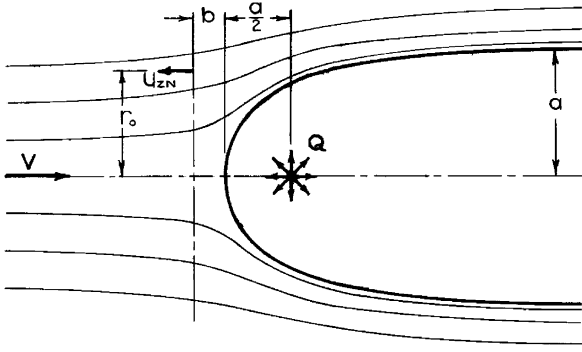


Fig. 16. A source representing the flow in the neighborhood of the nose of a nacelle.

If the streamlines near the nose of the nacelle are adequately represented by a single point source on the axis (Fig. 16), it is easily shown that

$$\bar{u}_{zN} = u_{zN}/V = -\frac{k/4}{[(r_0/a)^2 + k^2]^{3/2}}, \quad (6.2.3)$$

$$k = b/a + \frac{1}{2},$$

where a is the asymptotic radius of the source body and b the distance of the plane of the propeller ahead of the nose of the nacelle. Note that u_{zN} is negative and is indicated in the negative direction in Fig. 16. Eq. (6.2.2) may now be integrated and the radii of the blade elements of the free running propeller and of the propeller at the nacelle are related by the following:

$$(1 + \bar{u}_{z_0})(r^2 - r_0^2) = \frac{1}{2}a^2 \left[\frac{ka}{\sqrt{r_0^2 + k^2a^2}} - 1 \right]. \quad (6.2.4)$$

The resultant relative velocity at the blade element is given by Eq. (4.3.4) modified for the interference velocity u_{zN} due to the nacelle:

$$(U_0/V)^2 = (1 + \bar{u}_{z_0} + \bar{u}_{zN})^2 + (x_0/\lambda - \bar{u}_{\theta_0})^2, \quad (4.3.4a)$$

where $\bar{u}_{\theta_0}$ and $\bar{u}_{z_0}$ are given by Eqs. (4.3.2) and (4.3.3) and similarly the pitch angle of the relative wind is given by Eq. (4.3.5) suitably modified:

$$\tan \phi_0 = \frac{1 + \bar{u}_{z_0} + \bar{u}_{zN}}{x_0/\lambda - \bar{u}_{\theta_0}}. \quad (4.3.5a)$$

Lift coefficient and blade chord are then made such as to satisfy Eq. (4.5.2) and the blade angle is $\beta(x_0) = \alpha + \phi_0$.

Thrust and torque are then given by Eq. (5.2.2) and (5.2.3). The difference between the thrust and the thrust of the free running propeller is the force on the nacelle due to the presence of the propeller.

If the geometry of the free running propeller is not modified for the presence of the nacelle the loading will be increased, especially over the inner portions of the blades.

6.3. Propeller running in a wake

A well-developed wake is the consequence of the growth of a boundary layer or separated flow and is a region of reduced total pressure. As such, it has a very different effect on the performance of a propeller than does an irrotational velocity perturbation. Fig. 17 suggests the conditions for a propeller running in a uniform wake where there is a velocity deficiency of magnitude u_w from the free stream velocity V . Note that it is implied that u_w is in the opposite sense from u_z . The wake is assumed to be at the static pressure of the onset flow.

The total pressure in the ultimate slipstream is

$$p_\infty + \frac{\rho}{2}(V - u_w)^2 + \Delta p = p_\infty + \frac{\rho}{2}(V + u_z)^2, \quad (6.3.1)$$

where Δp is the static pressure increase through the impulse disc. Then

$$\Delta p/\rho = u_z(V + u_z/2) + u_w(V - u_w/2). \quad (6.3.2)$$

The thrust is

$$T = \Delta p S_0 = \rho S_0 [u_z(V + u_z/2) + u_w(V - u_w/2)]. \quad (6.3.3)$$

By the momentum theorem, Eq. (2.2.8), the thrust is

$$T = \rho S_0 (V + u_{z_0}) [(V + u_z) - (V - u_w)]. \quad (6.3.4)$$

Equating the two expressions for thrust, we find

$$u_{z_0} = \frac{1}{2}(u_z - u_w). \quad (6.3.5)$$

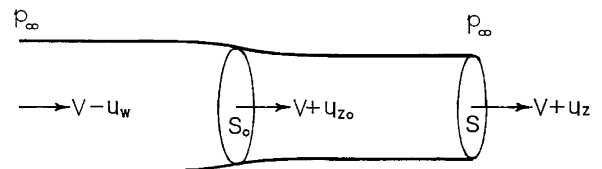


Fig. 17. Flow through a propeller running in a wake.

The velocity of the flow through the propeller is the mean of the remote upstream and downstream velocities as for the simple impulse disc (Appendix A).

The work done by the propeller is equal to the increase of kinetic energy in the slipstream:

$$dE/dt = \frac{1}{2}\rho S(V + u_z)[(V + u_z)^2 - (V - u_w)^2]. \quad (6.3.6)$$

The propulsive efficiency is

$$\eta = \frac{TV}{dE/dt} = \frac{1}{1 + \frac{1}{2}(u_z - u_w)/V}. \quad (6.3.7)$$

Denoting velocities divided by the free stream velocity with a bar, the thrust coefficient is, from Eq. (6.3.1),

$$K_T = 2[\bar{u}_z(1 + \bar{u}_z/2) + \bar{u}_w(1 - \bar{u}_w/2)]. \quad (6.3.8)$$

Solving this equation for $\bar{u}_z$ and substituting in Eq. (6.3.7), we have an expression for efficiency in terms of just the thrust coefficient and the wake velocity deficit $\bar{u}_w$:

$$\eta = \frac{2}{1 + \sqrt{1 + K_T - 2\bar{u}_w + \bar{u}_w^2} - \bar{u}_w}. \quad (6.3.9)$$

It is apparent that the effect of the velocity deficit (positive $\bar{u}_w$) of a wake flowing into a propeller is to *increase* the propulsive efficiency. This is in contrast with the effect of a reduction of velocity in an irrotational field which reduces efficiency (Section 6.1).

In principle it is even possible for the ideal efficiency to exceed 100%. This is not paradoxical. A propeller running in a wake is recovering energy previously lost to viscous effects. This effect is well known to naval architects who are usually dealing with propellers totally immersed in the very strong wake of the hull of a ship.

The treatments of an impulse disc in an accelerated irrotational field, Section 6.1, and in a wake, Section 6.3, provide an understanding of the qualitative effects of these perturbations, but in assuming a uniformly loaded disc in a uniform field, provide no computational procedure for a propeller in real circumstances. Qualitative understanding, however, can be a useful guide to choices of configuration.

In principle, the pusher propeller has significant advantages in potentially greater propulsive efficiency and in that it does not immerse the nacelle or fuselage in a drag-increasing slipstream. In practice, these advantages may be difficult to realize in

consequence of configurational complications, problems of engine cooling in the case of reciprocating engines, and propeller vibration induced by an unsymmetrical wake.

7. Regimes of operation of a propeller and a windmill

The vortex theory of propellers and the impulse disc simplified model (Appendix A) assume the formation of a definite slipstream flowing behind the propeller. This is the case in normal propulsion and for the windmill. However, a sequence of states of operation of airscrews may be discerned in which normal propulsion and the windmill are the two limiting modes. This sequence includes modes of operation where no true slipstream is formed.

Consider a propeller continuing to produce thrust, but at reduced forward velocity until it reaches the static case where it ceases to move through the air. In this case, it continues to produce a flow through the disc and a well-defined slipstream. Now assume it to move backward. At some velocity, the condition is attained where there is no net velocity through the propeller disc and no slipstream can form. Before this condition is reached a recirculating flow in the form of a vortex ring will occur. At increasing reverse velocities the flow through the disc changes direction and a turbulent wake is formed. At still higher velocity the thrust reverses direction and the airscrew becomes a windmill with a well-formed slipstream.

7.1. Flow through the disc when a well developed slipstream is formed

A general description of the states of operation can be formed with the aid of the impulse disc analysis. The thrust of the impulse disc is, from Appendix A, Eq. (A.7),

$$T = 2\rho u_{z_0}(V + u_{z_0})S, \quad (7.1.1)$$

where S is the disc area.

Since the quantities in this equation are changing sign, it is necessary to establish appropriate conventions. Let the direction of V , the onset flow, be always positive and a force on the propeller be considered positive when it is in the negative direction, as it is in normal propeller operation. Eq. (7.1.1) is then applicable in both the propeller mode and the windmill mode, but in the latter case T and u_{z_0} are both negative.

Define two dimensionless parameters to characterize the onset velocity and the velocity through the disc:

$$m = \frac{V^2}{T/2\rho S}, \quad (7.1.2)$$

$$M = \frac{(V + u_{z_0})^2}{T/2\rho S}. \quad (7.1.3)$$

From Eqs. (7.1.1) and (7.1.2) we have

$$T/2\rho S = V^2/m = u_{z_0}(V + u_{z_0}) \quad (7.1.4)$$

and from Eqs. (7.1.1) and (7.1.3),

$$T/2\rho S = (V + u_{z_0})^2/M = u_{z_0}(V + u_{z_0}). \quad (7.1.5)$$

Eliminating u_{z_0} from the last two equations, we have

$$m = (M - 1)^2/M. \quad (7.1.6)$$

This equation characterizes the velocity relations in the propeller and windmill modes where a well-developed slipstream exists and is shown in Fig. 18 as the solid lines. The equation expresses the velocity relations deduced from the simple impulse disc representation of a propeller, but the resulting graphic representation is a good qualitative picture of the regimes of operation of a propeller and a windmill. In the interval $-1 < M < 1$ Eq. (7.1.6) is not physically meaningful.

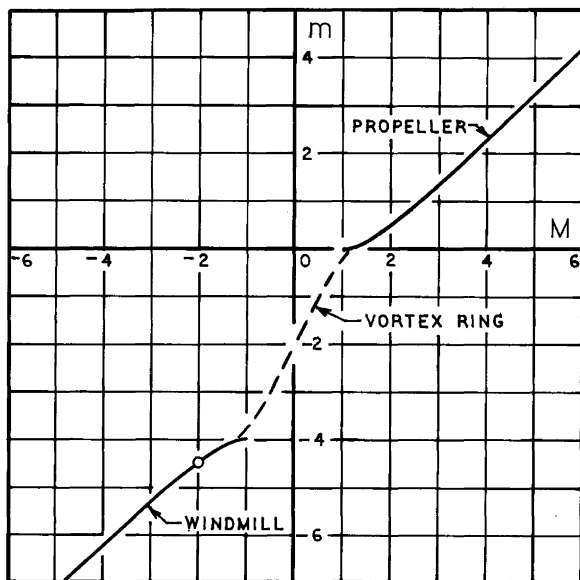


Fig. 18. Regimes of operation of an airscrew.

7.2. The vortex ring state

As the mean velocity through the propeller approaches zero, it enters a state where there is a recirculating flow through the disc resembling a vortex ring. The flow is complex and the details depend on the geometry of the propeller as well as the operational conditions. The thrust must accompany a transfer of momentum to the surrounding air and this can only take place by a turbulent mixing process. No theoretical model of this condition has been put forth. As the mean velocity through the propeller becomes negative (that is, of opposite sense to the onset flow) the flow becomes turbulent, perhaps similar to the wake flow behind an impervious disc. The vortex ring and turbulent states are indicated in the figure as a broken line. Experiments with propellers have provided data points for this regime, but the data is scattered.

There is also evidence that the vortex ring state tends to be unstable. It is well known that vertical descent of a helicopter with partial power, which puts the rotor in the vortex ring state, is a dangerously unstable maneuver where there may be sudden changes in the rate of descent.

7.3. The windmill

The aerodynamics of a windmill differs from that of a propeller in that the velocity through the disc is retarded rather than accelerated, but a well-developed slipstream exists behind a properly designed windmill. An efficient windmill will have a trailing vortex sheet in the form of a uniformly translating helicoid just as does a propeller. The design of a windmill of maximum efficiency can be carried out by the method set forth for a propeller in Section 5.1 with little modification. The essential difference is that the displacement velocity $\bar{w}$ must be assigned a negative value. The accompanying figure shows the components of velocity relative to a blade element. The induced axial velocity at the disc, u_{z_0} , is shown as a vector in the negative direction (opposite to V) as it must be (Fig. 19).

By the use of the impulse disc model it is a simple matter to derive a limiting value of the energy that can be recovered by a windmill. In converting the kinetic energy of the wind to mechanical power the velocity of the flow through the disc is reduced, thereby reducing the available kinetic energy. Consequently, it is evident that the maximum available energy must be less than the kinetic energy that would flow through the disc area in the absence of any energy conversion.

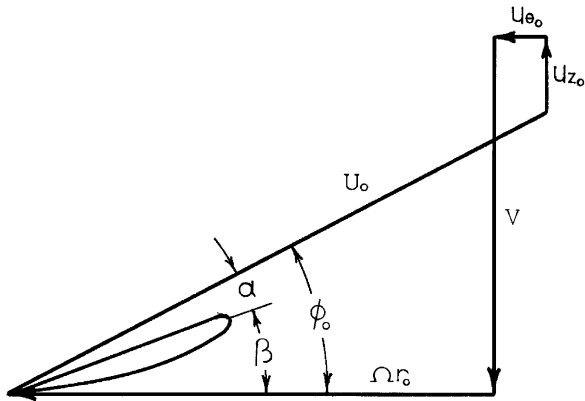


Fig. 19. Velocities at a blade element of a windmill.

While the thrust of a propeller is of primary importance, the axial force on the windmill is of only secondary interest. The energy converted by the impulse disc is equal to the thrust times the velocity through the disc. From Eq. (7.1.1) the energy is then

$$P = -2\rho u_{z_0}(V + u_{z_0})^2 S. \quad (7.3.1)$$

Differentiating with respect to u_{z_0} , it is found that the power is maximized when $u_{z_0}/V = -1/3$. The ratio of the converted energy to the kinetic energy flowing unimpeded through the disc is

$$\begin{aligned} P/P_0 &= \frac{-2\rho u_{z_0}(V + u_{z_0})^2 S}{\frac{1}{2}\rho V^3 S} \\ &= -4(u_{z_0}/V)(1 + u_{z_0}/V)^2, \end{aligned} \quad (7.3.2)$$

when $u_{z_0}/V = -1/3$ this is equal to 16/27, which is to say that no more than 59% of the energy that would flow unimpeded through the disc area is recoverable. The maximum recovery point corresponds to $m = -4.50$ and is indicated in Fig. 18 by the small circle on the windmill curve.

Appendix A

A.1. The impulse disc

There are several idealizations of the propeller which, by their relative simplicity, permit the derivation of expressions for limiting efficiency and provide some information about the velocity field of the propeller. The simplest of these is the impulse disc representation in which the propeller is idealized as a disc through which fluid flows freely and has a pressure increase imparted to it. The pressure rise is assumed to be uniform over the disc. There are

assumed to be no radial nor tangential forces on the fluid and continuity is preserved through the disc. The impulse disc representation of the propeller is often treated in an essentially one-dimensional analysis, which is not only an inaccurate application of the momentum principle, but glosses over a number of difficulties in what appears to be a very simple analysis. Examined in detail, it is perhaps not so simple, there being subtle problems, for instance, about the nature of the flow at the edge of the disc. Nonetheless, it establishes an upper bound for efficiency and provides a framework for some basic ideas about propulsion systems.

While very useful for the establishment of some simple relations, it should be recognized that the impulse disc is not a true limiting case for the screw propeller since all rotational effects are ignored.

At the disc there is necessarily a radial component of velocity so that streamlines are converging toward the axis. The pressure rise represents a force on the fluid normal to the disc and not in the direction of the streamlines. This is not paradoxical but merely means that the disc is a representation of a mechanism of unspecified type having certain characteristics, one of which is the absence of radial forces (Fig. 20).

The velocity components u_r and u_z are taken as perturbations from a uniform stream velocity V relative to the disc. The slipstream velocity u_z is taken far downstream where streamlines are parallel to the axis and the head is

$$H = p_\infty/\rho + V^2/2 + \Delta p/\rho = p/\rho + \frac{1}{2}(V + u_z)^2, \quad (A.1)$$

hence

$$\Delta p = p - p_\infty + \rho u_z(V + u_z/2).$$

From the Euler equation it is evident that downstream where all streamlines are parallel to the axis $p - p_\infty = 0$. Then

$$\Delta p = \rho u_z(V + u_z/2) \quad (A.2)$$

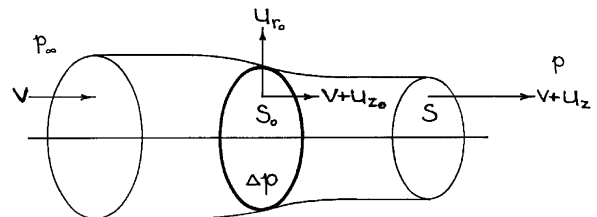


Fig. 20. The impulse disc.

and the thrust is just

$$T = \Delta p S_0 = \rho u_z (V + u_z/2) S_0. \quad (\text{A.3})$$

We also have, from the momentum Eq. (2.2.8)

$$T = \int_S \rho (V + u_z) u_z dS = \rho (V + u_z) u_z S. \quad (\text{A.4})$$

By continuity on a stream tube

$$(V + u_z) dS = (V + u_{z0}) dS_0.$$

Hence, Eq. (A.4) may be written in the alternative form

$$\begin{aligned} T &= \int_{S_0} \rho (V + u_{z0}) u_z dS_0 \\ &= \rho V u_z S_0 + \rho u_z \int_{S_0} u_{z0} dS_0. \end{aligned} \quad (\text{A.5})$$

This must be equal to Eq. (A.3) from which equality one finds

$$(1/S_0) \int_{S_0} u_{z0} dS_0 = u_z/2.$$

The left-hand side of this equation is just the average of u_{z0} over the disc:

$$u_{z0}(\text{avg}) = u_z/2. \quad (\text{A.6})$$

Thus, it is established that the *average* axial velocity at the actuator is the mean of the velocities far ahead and far behind.

In view of the integral form of the momentum equation, it cannot be concluded in general that u_{z0} is constant over the disc and equal to $u_z/2$, although the derivation is frequently presented in such a way that this conclusion is reached. For a lightly loaded impulse disc it can be shown from consideration of the vortex sheet shed from the edge of the disc that u_{z0} is constant and equal to $u_z/2$, hence the variation of u_{z0} over the disc is probably not great except near the edge of the disc unless the loading is heavy.

Eq. (A.3) is usefully expressed in terms of the velocity through the impulse disc:

$$T = 2\rho u_{z0} (V + u_{z0}). \quad (\text{A.7})$$

A.2. Efficiency

The efficiency of the impulse disc is easily evaluated in terms of conditions downstream

$$\begin{aligned} \eta &= \frac{TV}{dE/dt} = \frac{\rho (V + u_z) u_z VS}{\frac{1}{2} \rho (V + u_z) S [(V + u_z)^2 - V^2]} \\ &= 1/(1 + \frac{1}{2} u_z/V). \end{aligned} \quad (\text{A.8})$$

This may be expressed in terms of a thrust coefficient $K_T \equiv T/\frac{1}{2} \rho V^2 S_0$.

Observing Eq. (A.3), the thrust coefficient is

$$K_T = 2(u_z/V)(1 + \frac{1}{2} u_z/V). \quad (\text{A.9})$$

Consequently,

$$\eta = 2/(1 + \sqrt{1 + K_T}). \quad (\text{A.10})$$

Appendix B

B.1. The velocity induced by semi-infinite helicoidal vortex sheets

Consider a set of helicoidal vortex sheets symmetrically disposed about their axis springing from a plane normal to the axis:

- The axial and tangential velocities induced at the terminal plane are half of the induced velocities on the vortex sheets far from the plane of origin.
- *Corollary*—The induced velocity at the terminal edges of the vortex sheets is normal to the sheet.

The proof follows.

Consider the induced velocity u at a point $P(x_0, 0, 0)$ due to any single helical vortex element of the helicoidal sheet (Fig. 21).

The Biot–Savart law,

$$d\mathbf{u} = (\Gamma/2\pi a^3) d\mathbf{s} \times \mathbf{a}.$$

The equation of the helix is

$$z = \lambda(\theta - \theta_k), \quad r = \text{constant}.$$

Now

$$d\mathbf{s} = \mathbf{i} dx + \mathbf{j} dy + \mathbf{k} dz,$$

$$\mathbf{a} = \mathbf{i}(x_0 - x) - \mathbf{j}y - \mathbf{k}z,$$

where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are unit vectors in the x, y, z directions.

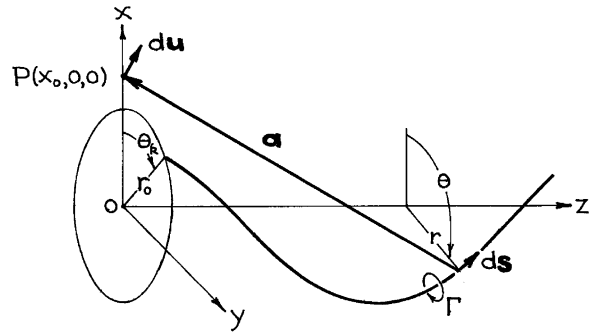


Fig. 21.

Therefore,

$$\mathbf{ds} \times \mathbf{a} = \mathbf{i}[y \, dz - z \, dy] + \mathbf{j}[z \, dx - (x_0 - x) \, dz] \\ + \mathbf{k}[-y \, dx - (x_0 - x) \, dy]$$

but

$$dx = -r \sin \theta \, d\theta,$$

$$dy = r \cos \theta \, d\theta,$$

$$dz = \lambda \, d\theta.$$

Consequently,

$$\mathbf{ds} \times \mathbf{a} = \mathbf{i}[\sin \theta - (\theta - \theta_k) \cos \theta] \lambda r \, d\theta \\ + \mathbf{j}[-(\theta - \theta_k) \sin \theta - x_0/r + \cos \theta] \lambda r \, d\theta \\ + \mathbf{k}[1 - (x_0/r) \cos \theta] r^2 \, d\theta$$

also

$$a^3 = [(x_0 - x)^2 + y^2 + z^2]^{3/2} \\ = [x_0^2 - 2x_0 r \cos \theta + r^2 + \lambda^2(\theta - \theta_k)^2]^{3/2}.$$

Therefore, by the Biot–Savart law,

$$du_x = (\Gamma/2\pi a^3)[\sin \theta - (\theta - \theta_k) \cos \theta] \lambda r \, d\theta,$$

$$du_y = (\Gamma/2\pi a^3)[-(\theta - \theta_k) \sin \theta \\ - x_0/r + \cos \theta] \lambda r \, d\theta,$$

$$du_z = (\Gamma/2\pi a^3)[1 - (x_0/r) \cos \theta] r^2 \, d\theta$$

and

$$du_x(\theta, \theta_k) = -du_x(-\theta, -\theta_k),$$

$$du_y(\theta, \theta_k) = +du_y(-\theta, -\theta_k),$$

$$du_z(\theta, \theta_k) = +du_z(-\theta, -\theta_k).$$

From these relations it is evident that the tangential u_y and axial u_z components induced by a symmetrically disposed set of infinite helical vortex lines are induced half by the part of the vortex on each side of the field point $P(x_0, 0, 0)$. The radial components induced by each semi-infinite part are of opposite sign and cancel. Consequently, the tangential and axial velocities induced by a semi-infinite set of vortex sheets at the terminal plane $z = 0$ are half the induced velocities on the sheets far from the plane.

Appendix C

C.1. The velocity field of a semi-infinite vortex cylinder

For problems concerning the flow in the vicinity of a propeller or of interference of a propeller with other parts of an aircraft or a ship, it is useful to have a reasonably simple model of the flow due to the action of a propeller. Such a model is the semi-infinite vortex cylinder, which may be taken as a simplified model of the field due to the shed vortex system of a propeller. The following discussion develops such a model.

Consider a closed vortex filament, not necessarily lying in a plane (Fig. 22).

The potential difference between any two points A and B along any path S in an irrotational field is the line integral

$$\varphi_B - \varphi_A = \int_A^B \mathbf{u} \cdot \mathbf{ds}.$$

If we consider a closed path around the vortex line which threads the loop, the potential increases by Γ , the strength of the vortex, upon the completion of each circuit. The potential is therefore multi-valued, increasing by Γ at each complete circuit (Fig. 23).

If, now, we imagine an arbitrary surface bounded by the vortex filament, the space that was doubly connected becomes singly connected. Letting there be a potential discontinuity through the surface equal to $-\Gamma$ at every point on the surface, the potential becomes single valued and the flow is unaltered since boundary conditions, including the gradient of the potential on either side of the surface, are unchanged. A surface across which there is a jump in potential is identically a doublet sheet whose strength at any point is equal to the potential difference. Consequently, the vortex of strength Γ is equivalent to an arbitrary

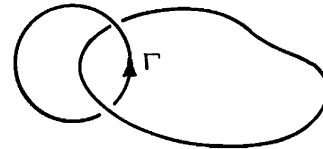


Fig. 22.



Fig. 23.

doublet sheet of uniform strength $-F$ bounded by the vortex line.

Now consider a cylindrical surface with uniformly distributed vorticity of strength γ per unit length, a vortex sheet that may be thought of as a uniform distribution of vortex rings. Let the cylinder extend indefinitely in one direction (Fig. 24).

Since a vortex ring is equivalent to a dipole sheet, the cylindrical vortex sheet is equivalent to a stack of circular dipole sheets. In the limit of continuous distribution along the axis, the individual potential jumps become a uniform potential gradient along the axis. This implies a uniform axial velocity within the cylinder. The potential gradient inside the cylinder is equal to the vorticity γ of the vortex sheet, giving a uniform velocity $u_i = \gamma$ inside the cylinder.

To this component of the field there must be added a sink disc of uniform strength covering the open end of the cylinder. The sink disc completes the field of the distributed dipoles. Its strength must be such that continuity is preserved across the disc when the sink disc is combined with the uniform axial velocity inside the cylinder. Letting the sink disc at the end of the cylinder have a strength $\mu = \gamma/2$, the velocity into the disc on each side is $u_s = \gamma/2$. On the outside the resultant axial velocity is $u = \gamma/2$. On the inside the velocity into the disc is $u_s = -\gamma/2$ which, combined with the uniform internal component $u_i = +\gamma$ gives a resultant $u = \gamma/2$, preserving continuity through the sheet.

Consequently, the complete field of the semi-infinite vortex cylinder is equivalent to a sink disc of strength $\mu = \gamma$ at the open end of the cylinder plus a uniform axial velocity everywhere within the cylinder superposed on the field of the sink disc. The axial velocity at the open end of the vortex cylinder is $\gamma/2$ and far down the cylinder as $z \rightarrow \infty$ the velocity is uniformly equal to γ inside and zero outside the cylinder.

The semi-infinite vortex cylinder may be taken to represent the velocity field of a lightly loaded impulse disc and may, to some degree of approximation, be taken to represent the field of more elaborate models of the flow around a propeller, especially far from the propeller.

Appendix D

D.1. The Kutta–Joukowski theorem in three-dimensional flow

In textbooks the Kutta–Joukowski theorem is derived for two-dimensional flow. More often than

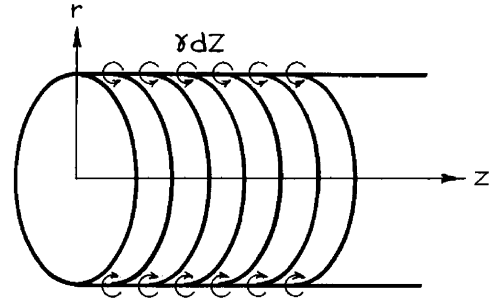


Fig. 24.

not it is applied to problems in three-dimensional flow without comment on nor justification of the generalization. A useful form of the theorem as applied to lifting surfaces in a three-dimensional flow is presented here with a demonstration of its validity.

Consider a rigid lifting surface in a steady irrotational flow. Any point on the surface may be considered to be the origin of orthogonal coordinates x, y, z where z is normal to the surface. The velocity on the upper (positive z) side of the surface is u_2 at an angle ϕ_2 to the x -axis and similarly the velocity on the lower side is u_1 at angle ϕ_1 to the x -axis (Fig. 25).

Under the stated conditions the pressure difference between the upper and lower surfaces of the sheet must be

$$\Delta p = \frac{1}{2} \rho (u_2^2 - u_1^2), \quad (\text{D.1})$$

where the resultant pressure is in the positive z direction. Now consider the surface as a bound vortex sheet. The following form of the Kutta–Joukowski theorem is proposed:

$$\Delta p = \rho \mathbf{u} \times \boldsymbol{\gamma}, \quad (\text{D.2})$$

where $\mathbf{u}$ is the vector mean of the upper and lower surface velocities u_1 and u_2 and $\boldsymbol{\gamma}$ the bound vortex density expressed as a vector. Eq. (D.2) may be written in the scalar form

$$\Delta p = \rho u_x \gamma_y - \rho u_y \gamma_x, \quad (\text{D.3})$$

where the components of the vortex density are

$$\begin{aligned} \gamma_x &= -u_2 \sin \phi_2 + u_1 \sin \phi_1, \\ \gamma_y &= u_2 \cos \phi_2 - u_1 \cos \phi_1 \end{aligned} \quad (\text{D.4})$$

and the components of the mean velocity at the lifting surface are

$$\begin{aligned} u_x &= \frac{1}{2} (u_1 \cos \phi_1 + u_2 \cos \phi_2), \\ u_y &= \frac{1}{2} (u_1 \sin \phi_1 + u_2 \sin \phi_2). \end{aligned} \quad (\text{D.5})$$

Substituting Eqs. (D.4) and (D.5) in (D.3), we recover the original expression (D.1). The identity of (D.1) and (D.2) is thereby demonstrated.

Appendix E

E.1. A Modification of Simpson's rule

Simpson's rule for numerical integration or determination of an area is accurate to the extent that the function over each successive pair of adjacent intervals is adequately represented by a second-order curve. This is very satisfactory for most smooth functions. However, if the terminus of the curve is parabolic or elliptical in character, i.e. if the curve is like $\sqrt{1-x}$ with the slope becoming infinite at the end where $x \rightarrow 1$, the second-order curve fit is poor and the integration for that part of the function (the first or last two intervals) is seriously underestimated. A simple modification of Simpson's rule minimizes the inaccuracy of integration of functions of this type (Fig. 26).

The figure represents such a parabolic function. The second ordinate from the end is f_{-2} , the next is $f_{-1} = \frac{1}{2}\sqrt{2}f_{-2}$ and the last ordinate f_0 is zero. By Simpson's rule, the area under the curve over the last two intervals is

$$\begin{aligned}\Delta A &= \frac{h}{3}(f_{-2} + 4f_{-1} + f_0) = \frac{h}{3}(f_{-2} + 2\sqrt{2}f_{-2} + 0) \\ &= \frac{h}{3}(1 + 2\sqrt{2})f_{-2} = 1.276hf_{-2}.\end{aligned}$$

The correct area for this parabolic segment of the curve is

$$\Delta A = \frac{2}{3}(2h)f_{-2} = \frac{4}{3}hf_{-2}.$$

The Simpson approximation for the area in the last two intervals is about 4.3% too low. For a better approximation, replace the multiplier 4 on

the middle term by a different value m :

$$\Delta A = \frac{h}{3}(f_{-2} + mf_{-1} + f_0),$$

$$\frac{4}{3}hf_{-2} = \frac{h}{3}\left(f_{-2} + m\frac{1}{2}\sqrt{2}f_{-2} + 0\right),$$

which gives $m = 3\sqrt{2} = 4.243$.

Consequently, for integrations terminating with a parabolic or elliptical character, the Simpson multiplier 4 in the next to last term should be replaced by 4.243. The last term is, of course, $1 \times 0 = 0$.

The modified rule was used in the computation of the mass transport factor κ (Table 2).

E.2. An example of a simple application

Compute the area of a quarter circle, $y = \sqrt{1-x^2}$, $0 \leq x \leq 1$ by Simpson's rule using only four intervals, $h = .25$ (Fig. 27).

x	y	Simpson's rule		Modified rule	
		m	my	m	my
0	1.0000	1	1.0000	1	1.0000
.25	.9682	4	3.8730	4	3.8730
.50	.8661	2	1.7321	2	1.7321
.75	.6614	4	2.6458	4.243	2.8065
1.00	0	1	0	1	0
			$\Sigma my = 9.2509$	$\Sigma my = 9.4116$	

$$A = \frac{.25}{3} \Sigma my = .7709, \quad \text{by Simpson's rule.}$$

$$A = \frac{.25}{3} \Sigma my = .7843, \quad \text{by modified rule.}$$

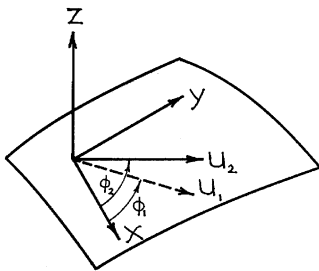


Fig. 25.

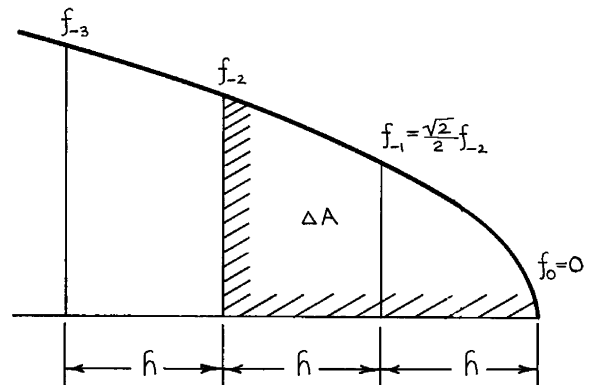


Fig. 26.

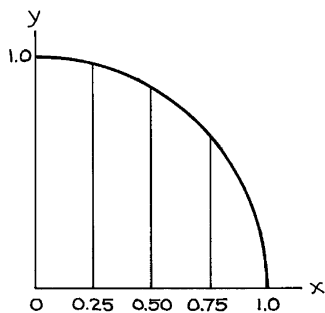


Fig. 27.

The true area is $\pi/4 = .7854$. The standard Simpson's rule results in a value that is 1.8% low. The modified Simpson's rule is only about .1% low, a remarkably accurate result where we are using a relatively crude computation using only five ordinates.

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